Amine

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Amine

Reactions of Primary Arylamines with Nitrous acid.

Reaction of Amines with Nitrous Acid

Nitrous acid (HONO) is a weak, unstable acid. It is always prepared in situ, usually by treating sodium nitrite ($NaNO_2$) with an aqueous solution of a strong acid:

$HCl_{(aq)} + NaNO_{2(aq)} \rightarrow HONO_{(aq)} + NaCl_{(aq)}$

$H_2SO_4 + 2NaNO_2(aq) \rightarrow 2HONO_{(aq)} + Na_2SO_{4(aq)}$

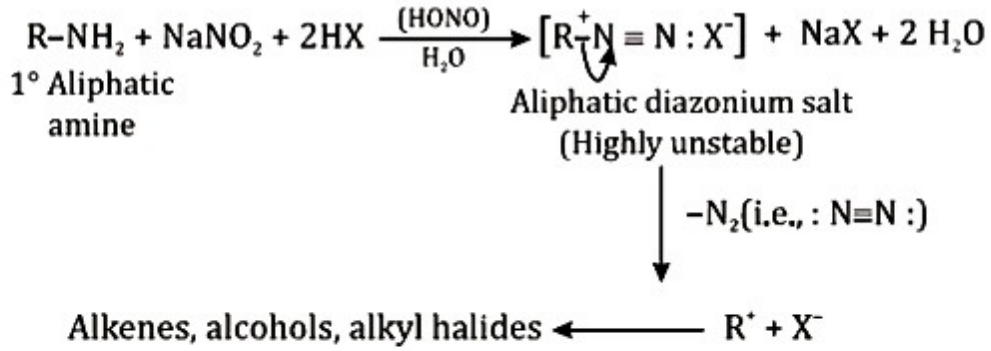
Reactions of Primary Aliphatic Amines with Nitrous acid

Primary aliphatic amines react with nitrous acid through a reaction called diazotization to yield highly unstable aliphatic **diazonium salts.**

At low temperatures, aliphatic diazonium salts decompose spontaneously by losing nitrogen to form carbocations.

Carbocations go on to produce mixture of alkenes, alcohols, and alkyl halides by removal of a proton, reaction with H_2O, and reaction with X–

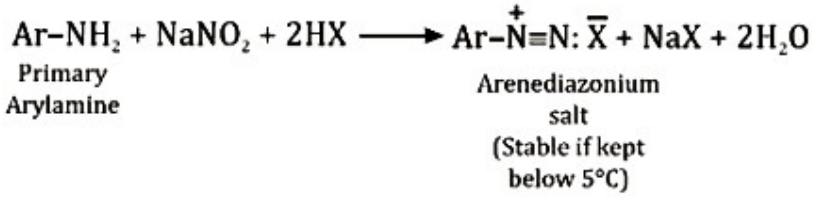
General Reaction



Diazotizations of primary aliphatic amines are little synthetic importance because they yields such a complex mixture of products. Diazotizations of primary aliphatic amines are used in some analytical procedures, however, because the evolution of nitrogen in quantitative. They can also be used to generate and thus study the behavior of carbocations in water, acetic acid, and other solvents.

Arenediazonium Salts

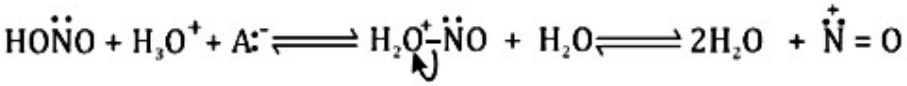
- The most important reaction amines with nitrous acid, by far, is the reaction of primary arylamines.
- Primary arylamines react with nitrous acid to give arenediazonium salts.
- Arenediazonium salts are unstable, they are still far more stable than aliphatic diazonium salts.
- They do not decompose at an appreciable rate in solution when the temperature of the reaction mixture is kept below 5°C.



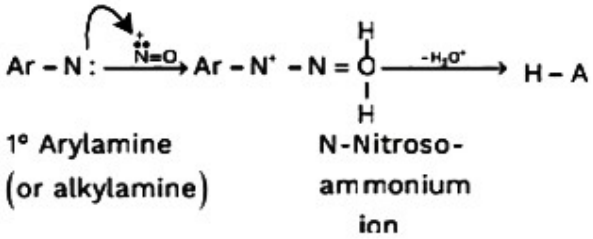
A Mechanism for the reaction

Diazotization of a primary amine takes place through a series of steps.

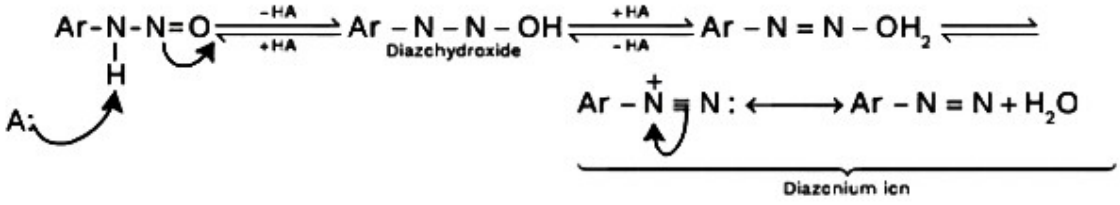
Diazotization



In the presence of strong acid, nitrous acid dissociates to produce +NO ions. These ions then react with the nitrogen of the amine to form an unstable N-nitrosoammonium ion as an intermediate.

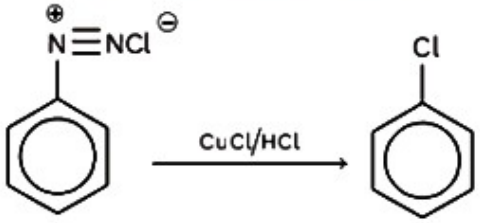


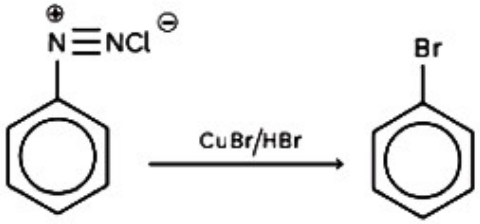
This intermediate then loses a proton to form an N-nitrosoamine, which in turn, tautomerizes to a diazohydroxide in a reaction that is similar to keto-enol tautomerization.



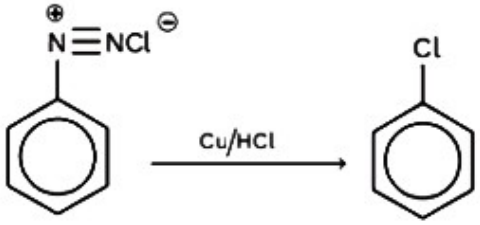
Diazotization reaction of primary arylamines are of considerable synthetic impotence because the diazonium group, $-\overset{+}{N}\equiv N:$, can be replaced by a variety of other functional groups.

Chemical reactions of benzene diazonium chloride

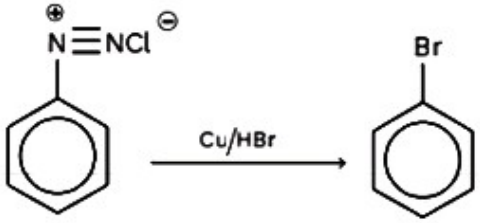
1. 

2. 

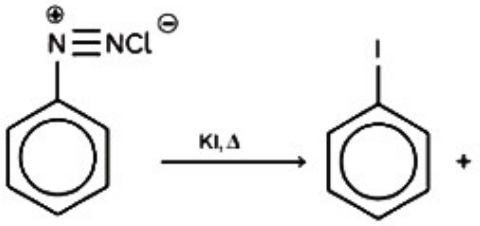
(Sandmeyer reaction)

3. 

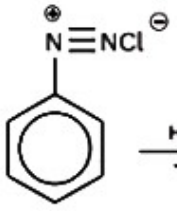
(Gattemann reaction)

4. 

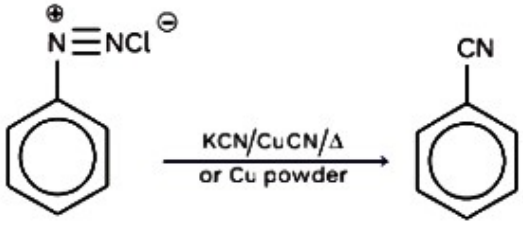
(Gattemann reaction)

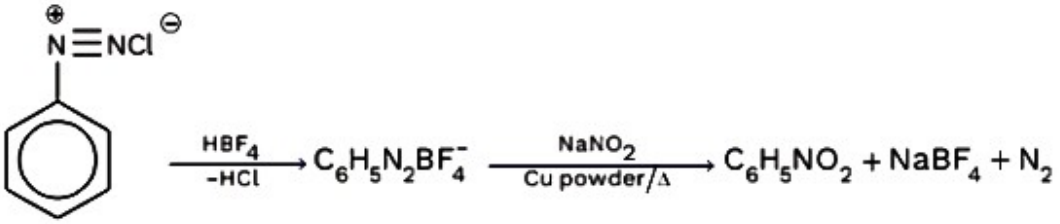
5. 

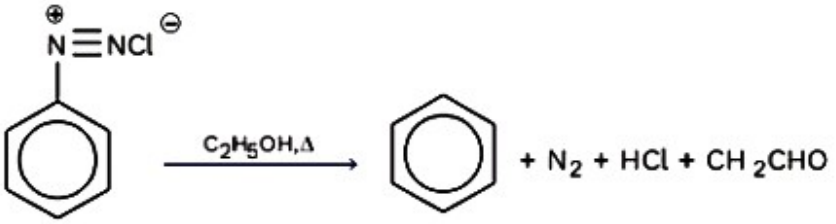
$+ N_2 + KCl$

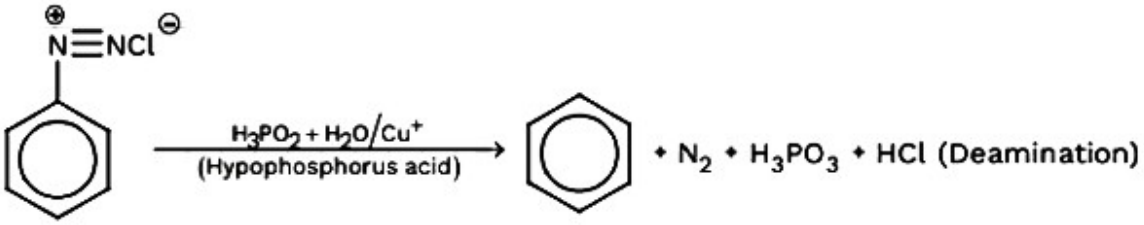
6. 

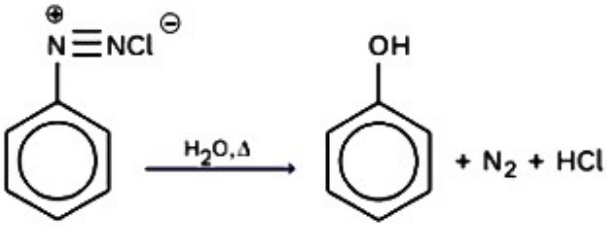
$C_6H_5N_2^{\oplus}BF_4^{\ominus} \xrightarrow{\Delta} C_6H_5 - F + BF_3 + N_2$ (Balz-Schiemann reaction)

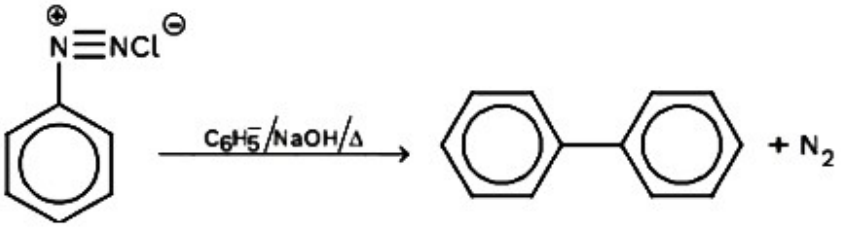
7. 

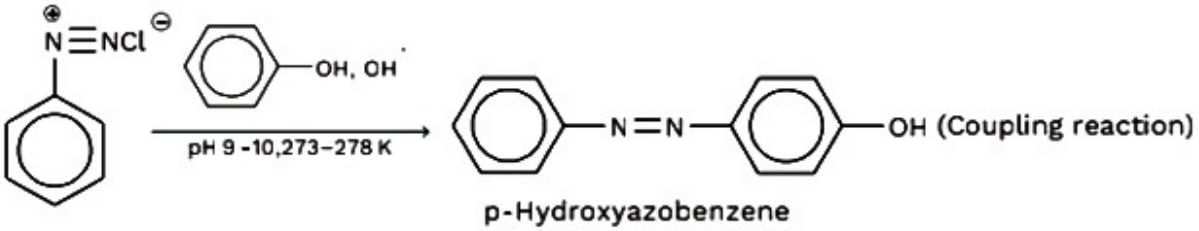
8. 

9. 

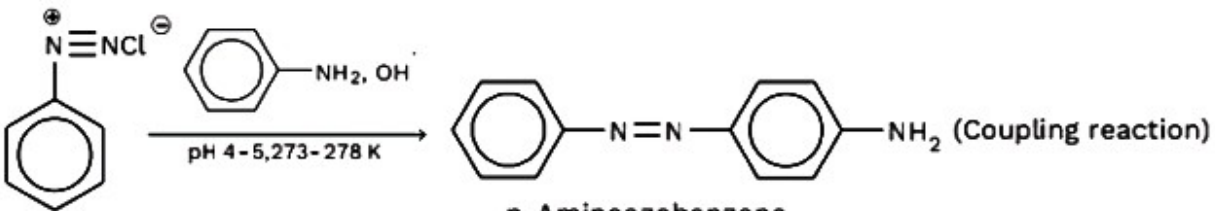
10. 

11. 

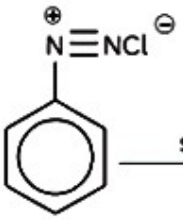
12. 

13. 

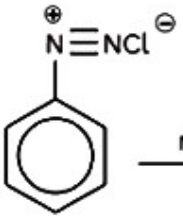
p-Hydroxyazobenzene

14. 

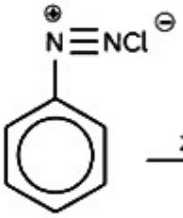
p-Aminoazobenzene
(yellow dye)

15. 

$\xrightarrow{SnCl_2/HCl} C_6H_5NHNH_2, HCl$

16. 

$\xrightarrow{Na_2SO_3/HCl} C_6H_5NHNH_2$

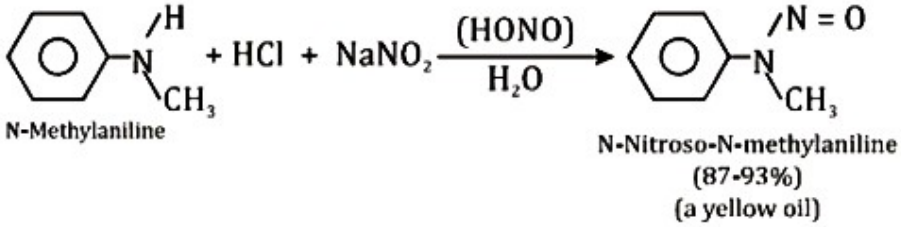
17. 

$\xrightarrow{Zn/HCl} [C_6H_5NHNH_2] \xrightarrow{Zn/HCl} C_6H_5NH_2 + NH_3$

Reactions of Secondary Amines with Nitrous Acid

Secondary amines – both aryl and alkyl – react with nitrous acid to yield N–nitrosoamines. N-Nitrosoamines usually separate from the reaction mixture as oily yellow liquids:

$$\underset{\text{Dimethylamine}}{(CH_3)_2NH} + HCl + NaNO_2 \xrightarrow[H_2O]{(HONO)} \underset{\substack{\text{N-Nitrosodimethylamine}\\\text{(a yellow oil)}}}{(CH_3)_2N{-}N{=}O}$$



N-Methylaniline

N-Nitroso-N-methylaniline
(87-93%)
(a yellow oil)

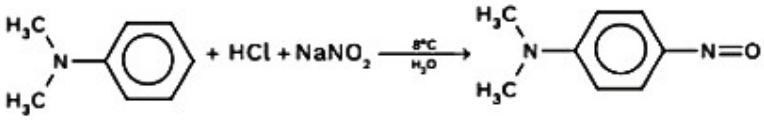
Reactions of Tertiary Amines with Nitrous acid.

When a tetriary aliphatic amine is mixed with nitrous acid, an equilibrium is estabilised among the tertiary amine, its salt, and an N-nitrosoammonium compound:

$$\underset{\substack{\text{Tertiary aliphatic}\\\text{amine}}}{2\,R_3N{:}} + HX + NaNO_2 \underset{+HA}{\overset{-HA}{\rightleftharpoons}} \underset{\text{Amine Salt}}{R_3\overset{+}{N}H\,\bar{X}} + \underset{\substack{\text{N-Nitrosoammonium}\\\text{Compound}}}{R_3\overset{+}{N}{-}N{=}O\bar{X}}$$

While N nitrosoammonium compounds are stable at low temperatures, at higher temperatures and in aqueous acid they decompose to produce aldehydes or ketones.
Tertiary arylamines react with nitrous acid to form C-nitroso aromatic compounds. Nitrosation takes place almost exclusively at the para position if it is open and, if not, at the ortho position. The reaction is another example of electrophilic aromatic substitution.

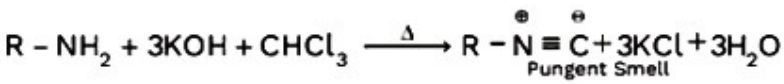
Specific Example



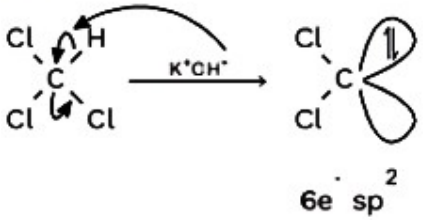
N-Nitroso-N,N-dimethylaniline
(80-90%)

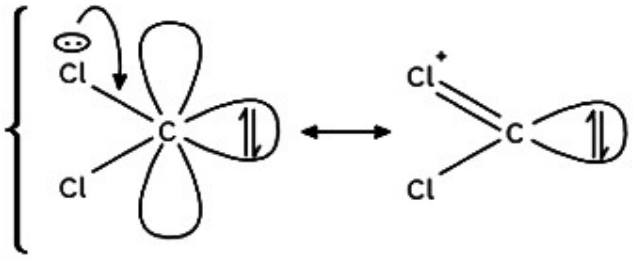
CARBYL AMINE TEST OR ISOCYNIDE TEST

1° amine → Isocynide

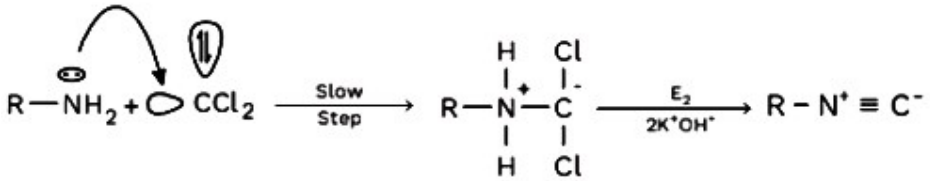


Mechanism:





Singlet Carbene



Characteristics of Reaction

(i) In isocynide test 1° amine convert in isocynide in presence of basic medium.

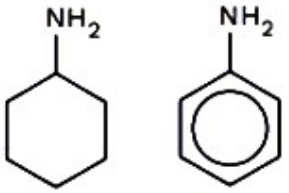
(ii) Carbene and carbanion formed as intermediate.

(iii) Carbene formation is (1, 1) Elimination or a-Elimination.

(iv) Singlet carbene is a resonance stabilised electrophile (Poor electrophile)

(v) More electron donor amine, more the rate of Reaction, more collision with carbene.

Q75



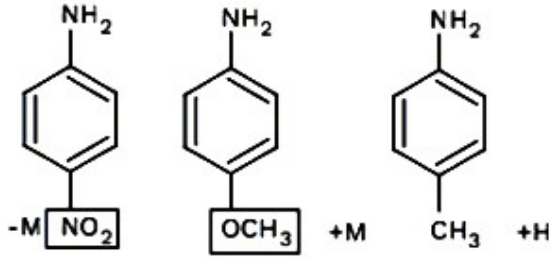
(I) (II)

Find out rate of Reaction.

Sol: **I > II**

(In case of 'II' amine lone involved in resonance with benzene).

Q76 **Find out rate of reaction**



(1) (2) (3)

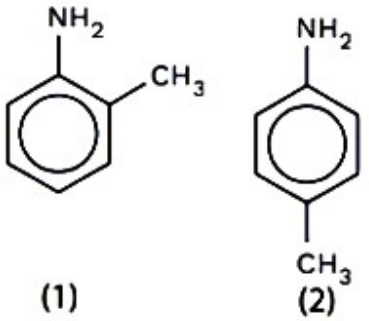
Sol: **2 > 3 > 1**

$-NO_2$ – M effect

$-OCH_3$ + M effect

$-CH_3$ + H effect

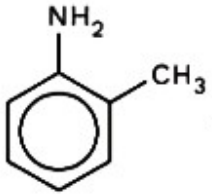
Q77 **Find out rate of reaction**



(1) (2)

Sol: **1 > 2**

(More electron donor benzene, higher rate of reaction)



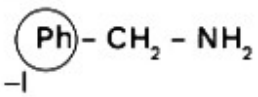
+I / +H More + I effect at ortho position

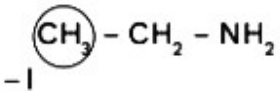
Q78 **Find out rate of reaction**

$CH_3 - CH_2 - NH_2$ $Ph - CH_2 - NH_2$

(1) **(2)**

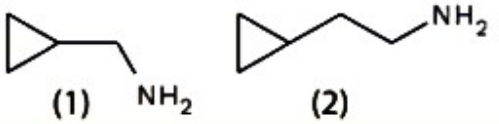
Sol: **1 > 2**





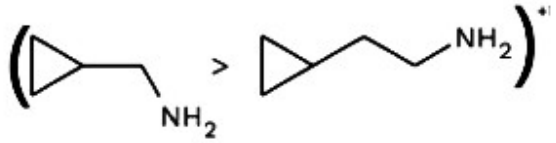
2>1

Q79 **Find out rate of reaction**

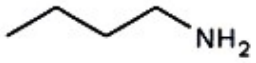


(1) (2)

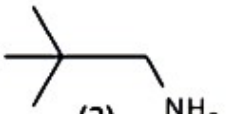
Sol: **1 > 2**



Q80 **Find out rate of reaction**

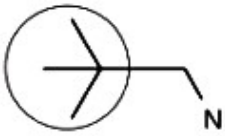


(1)



(2)

Sol: **2 > 1**



NH_2 Bigger size group, Higher + I effects

Advance Isocynide Test

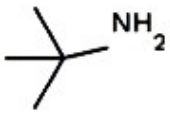
$$Ph-NH_2 \xrightarrow[+KOH/\Delta]{CHCl_3} Ph-\overset{\oplus}{N}\equiv C^{\ominus} \xrightarrow{H_2\overset{\bullet}{O}} A+B \rightarrow \text{Tollens Positive Test}$$

Mechanism :

$$Ph-\overset{\oplus}{N}\equiv \overset{\ominus}{C} \xrightarrow{H\,O^{\oplus}} Ph-NH_2 + H-C-OH$$

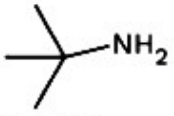
$$H-\overset{O}{\overset{\|}{C}}-OH \longrightarrow \text{aldehyde group, positive Test.}$$

Q81 **How many compounds may give isocynide test?**

1. CH_3-NH_2
2. $CH_3-NH-CH_3$
3. $H_3C-N(CH_3)-CH_3$
4. 
5. $Ph-NH_2$
6. $Ph-\overset{\oplus}{N}H_3\,Cl^{\ominus}$
7. $Ph-\overset{O}{\overset{\|}{C}}-NH_2$
8. $(Ph)_2^{\oplus}NHCl^{\ominus}$

Sol: **3**

CH_3-NH_2



Ph-NH_2

Q82 **Stability order of following singlet halocarbene is**

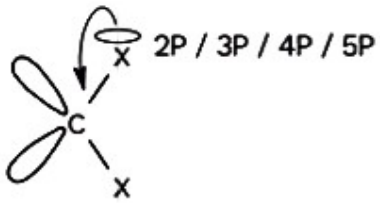
(A) $CF_2 > CCl_2 > CBr_2 > Cl_2$

(B) $Cl_2 > CBr_2 < CCl_2 > CF_2$

(C) $CCl_2 > CF_2 < CBr_2 > Cl_2$

(D) $CF_2 > Cl_2 > CCl_2 > CBr_2$

Sol: **A**



Lone Pair in Resonance with vacant orbital

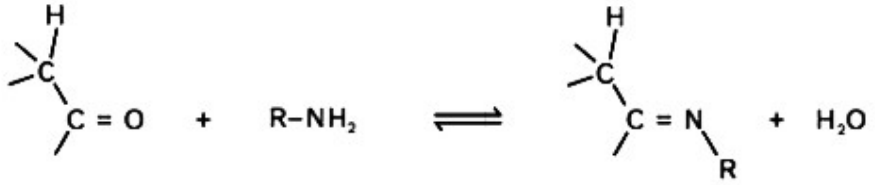
Smaller size, Higher tendered to delocalise.

Nitrogen Derivatives via Nucleophilic Mechanism

Reactions of Aldehydes and Ketones with Nitrogen Nucleophiles

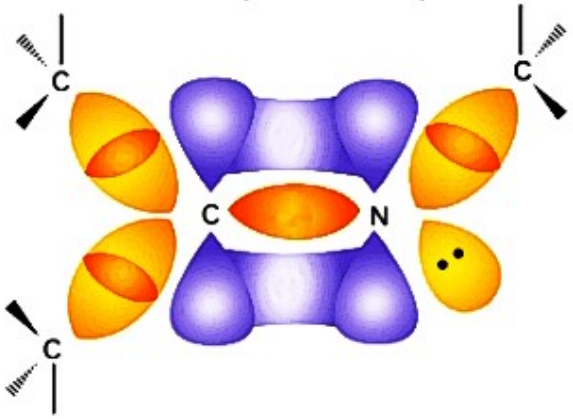
Reaction with primary amines

Aldehydes and ketones react with a primary amine (RNH_2) to form an imine. An imine is a compound with a carbon-nitrogen double bond. The imine obtained from the reaction of a carbonyl compound and a primary amine is often called a Schiff base.



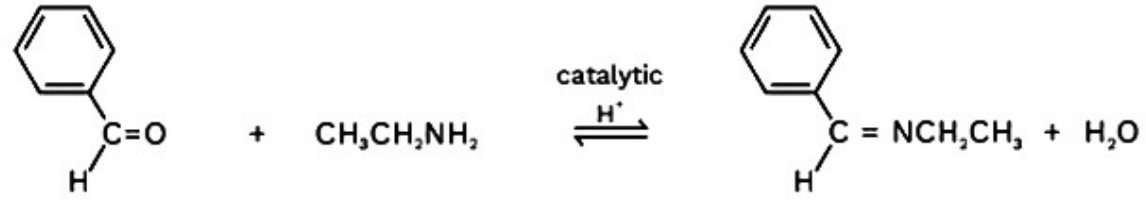
an aldehyde or a ketone — a primary amine — an imine a Schiff base

The orbital model of a C=N group is similar to the orbital model of a C=O group. The imine nitrogen is sp^2 hybridized. One of its sp^2 orbitals forms a s bond with the imine carbon, one forms a sigma bond with a substituent, and the third contains a lone pair. The p orbital of nitrogen and the p orbital of carbon overlap to form a p bond.



Addition of Primary Amines

Aldehydes and ketones react with primary amines to form imines. The reaction requires a catalytic (small) amount of acid—we will see that the pH of the reaction mixture must be carefully controlled.

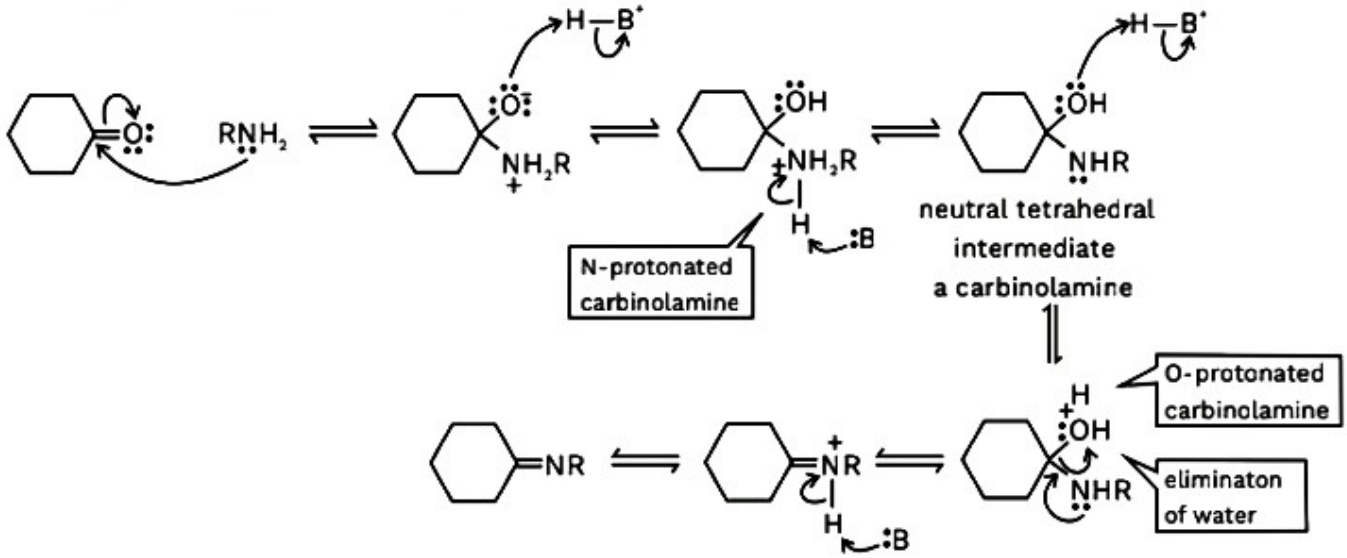


benzaldehyde
an aldehyde

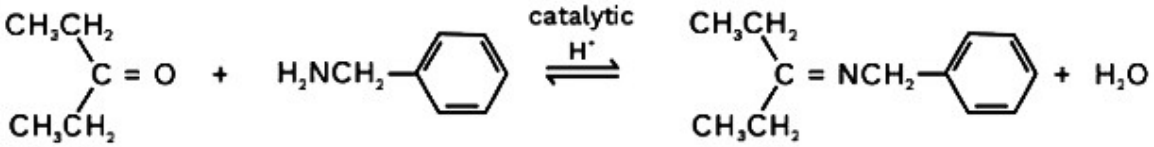
ethylamine
a primary amine

an imine

Mechanism for imine formation :



Example



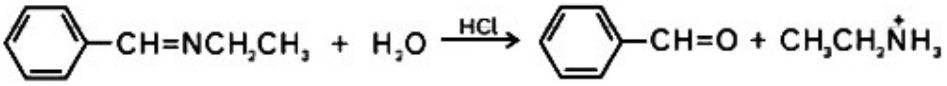
3-pentanone
a ketone

benzylamine
a primary amine

an imine

Reversibility in imine formation

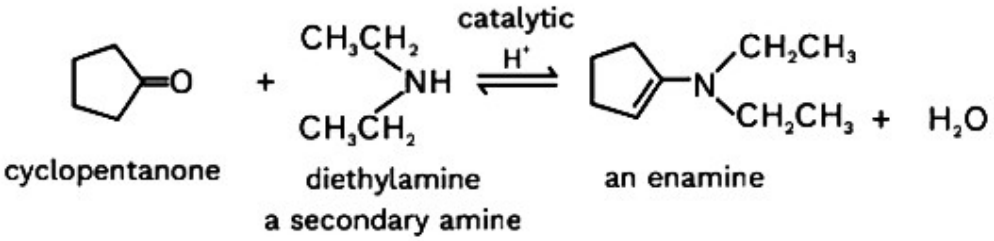
In acidic aqueous solutions, imines are hydrolysed back to the carbonyl compound and amine.

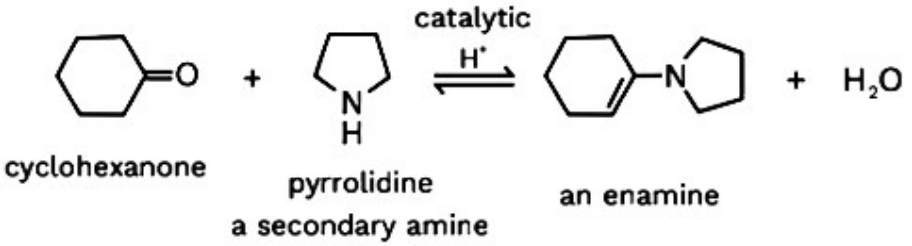


In an acidic solution, the amine is protonated and, therefore, is unable to react with the carbonyl compound to reform the imine.

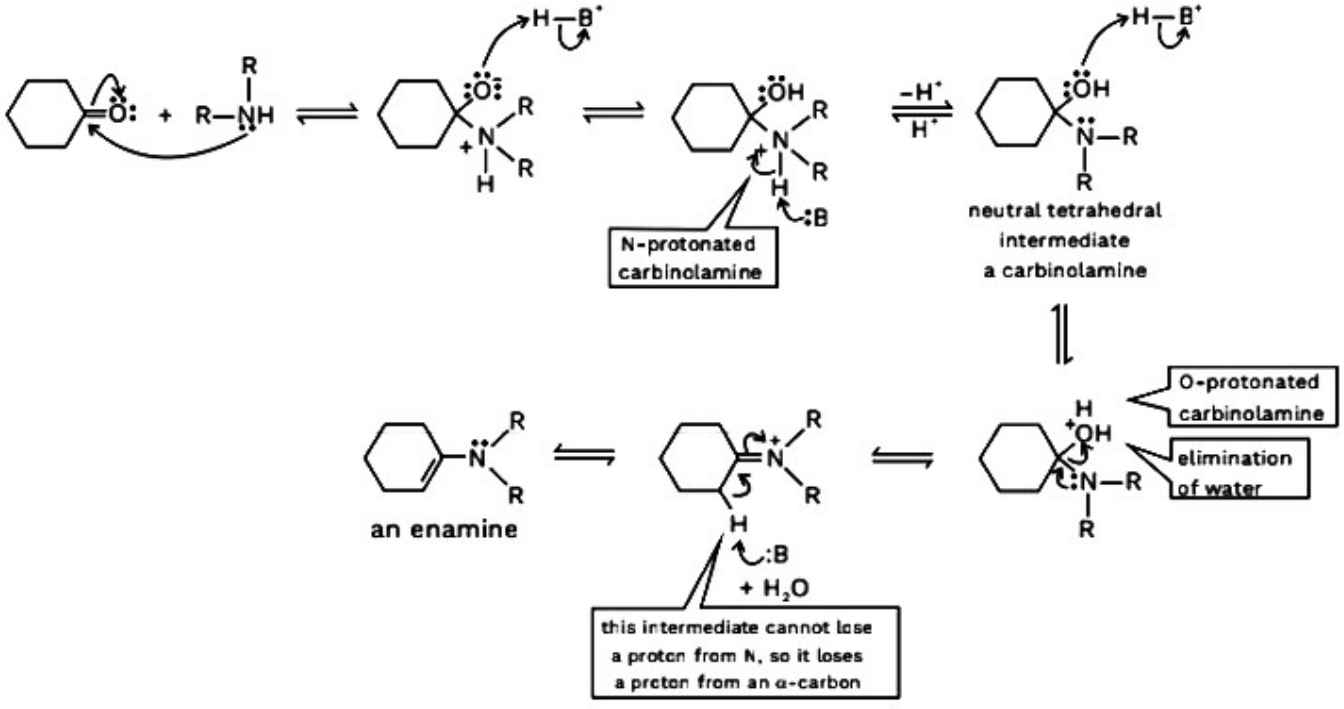
Addition of Secondary Amines

Aldehydes and ketones react with secondary amines to form enamines. Like imine formation, the reaction requires a trace amount of an acid catalyst.





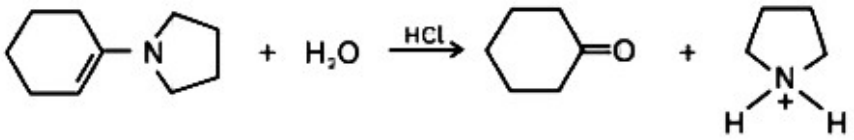
Mechanism for enamine formation:



The mechanism for enamine formation is exactly the same as that for imine formation, until the last step of the reaction. When a primary amine reacts with an aldehyde or a ketone, the protonated imine loses a proton from nitrogen in the last step of the reaction, forming a neutral imine. However, when the amine is secondary, the positively charged nitrogen is not bonded to a hydrogen. A stable neutral molecule is obtained by removing a proton from the -carbon of the compound derived from the carbonyl compound. An enamine is the result.

Note

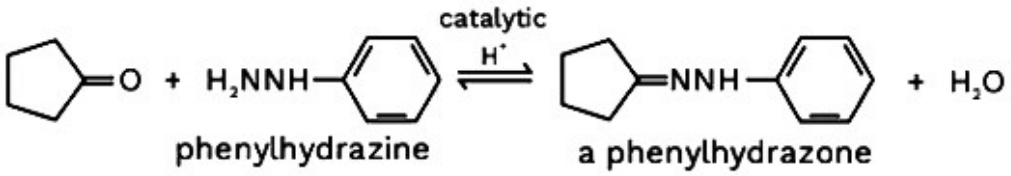
In aqueous acidic solutions, an enamine is hydrolysed back to the carbonyl compound and secondary amine, a reaction that is similar to the acid-catalysed hydrolysis of an imine back to the carbonyl compound and a primary amine.

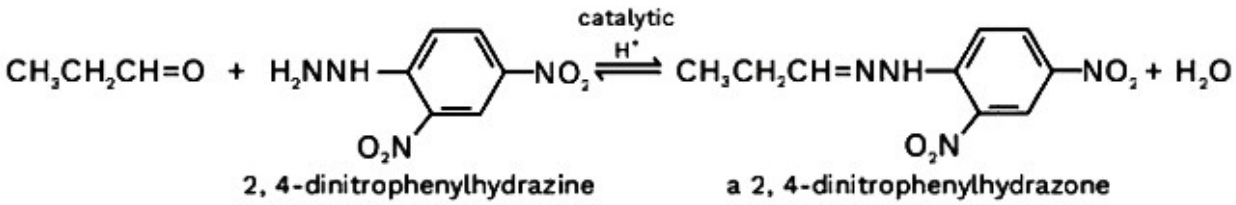


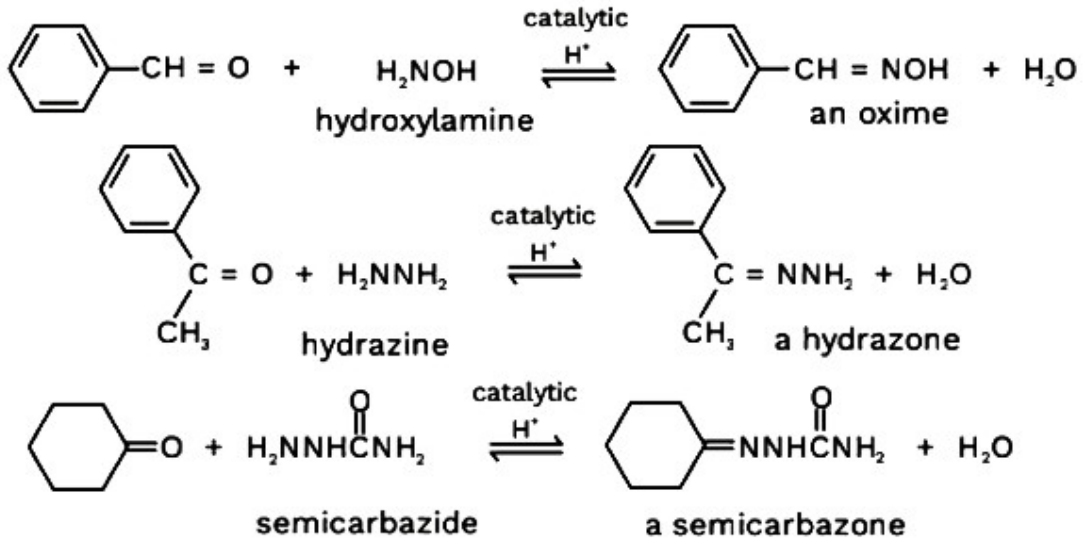
Formation of Derivatives

Compounds such as hydroxylamine (NH_2OH), hydrazine (NH_2NH_2), and semicarbazide ($NH_2NHCONH_2$) are similar to primary amines in that they all have an NH_2 group. Thus, like primary amines, they react with aldehydes and ketones to form imines—often called imine derivatives because the substituent attached to the imine nitrogen is not an R group. The imine obtained from the reaction with hydroxylamine is called an **oxime**, the imine obtained from the reaction with hydrazine is called a **hydrazone**, and the imine obtained from the reaction with semicarbazide is called a **semicarbazone**.

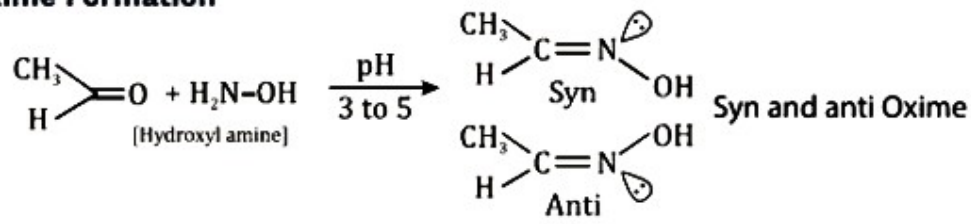
Phenyl-substituted hydrazines react with aldehydes and ketones to form phenyl-hydrazones.



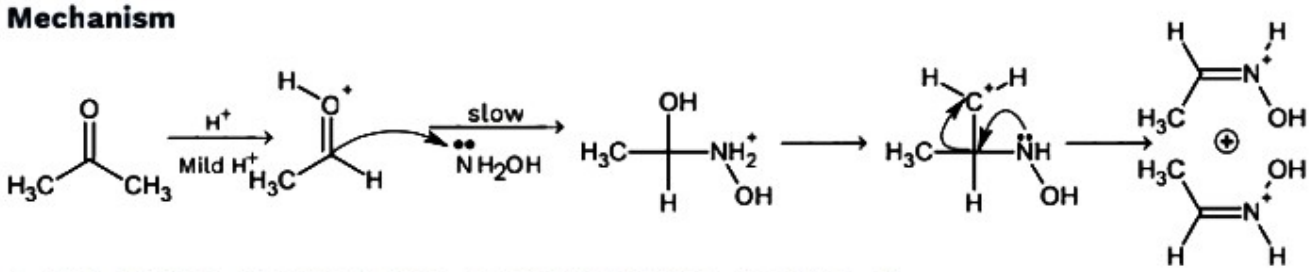




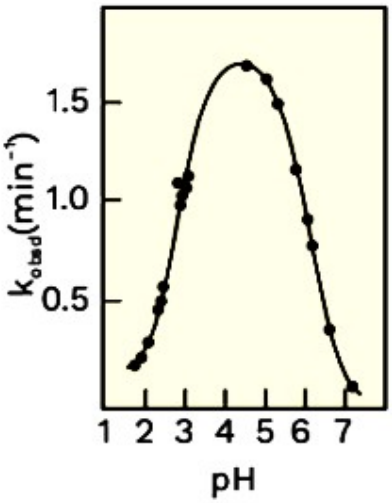
Oxime Formation



Mechanism



A plot of the observed rate constant for the reaction of acetone with hydroxylamine as a function of the pH of the reaction mixture is shown in Figure. This type of plot is called a pH-rate profile. The pH-rate profile in the figure is a bell-shaped curve with **the maximum rate occurring at about pH 4.5, 1.5** pH units below the pK_a of hydroxylamine (pK_a = 6.0). As the acidity increases below pH 4.5, the rate of the reaction decreases because more and more of the amine becomes protonated. As a result, less and less and less of the amine is present in the nucleophilic non protonated form. As the acidity decreases above pH 4.5, the rate decreases because less and less of the tetrahedral intermediate is present in the reactive protonated form.

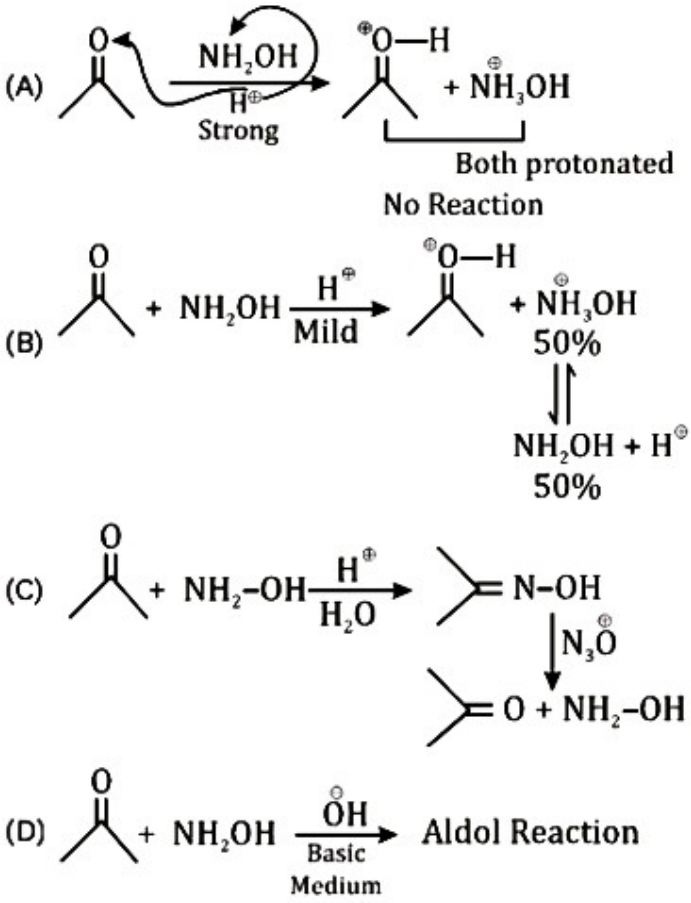


Q83 **Why Oxime formation is possible in mild acidic medium.**

(A) pH = 1 to 3 **(B) pH = 3 to 5**

(C) pH = 5 to 7 **(D) pH = 8 & above**

Sol:



The pK_a of protonated acetone is about −7.5 and the pK_a of protonated hydroxylamine is 6.0.

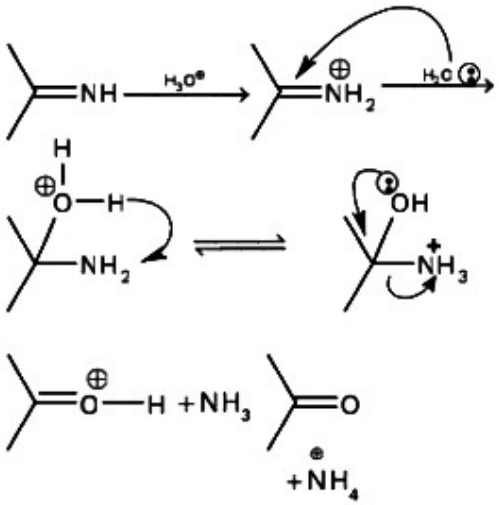
(a) In its reaction with hydroxylamine at pH = 4.5 (Figure), what fraction of acetone will be present in its protonated form?

(b) In its reaction with hydroxylamine at pH = 1.5, what fraction of acetone will be present in its acidic, protonated form?

(c) In its reaction with acetone at pH = 1.5 (Figure), what fraction of hydroxylamine will be present in its reactive basic form?

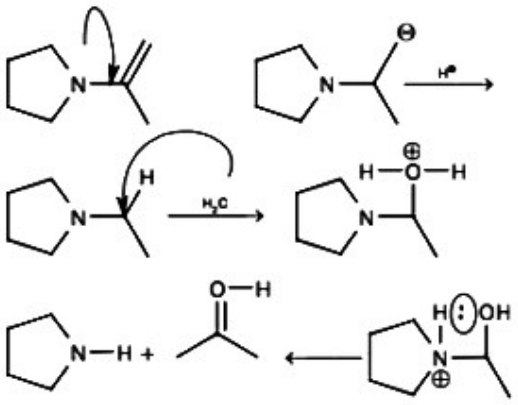
Q84 **Write the mechanism for the following reactions:**
The acid-catalyzed hydrolysis of an imine to a carbonyl compound and a primary amine

Sol:

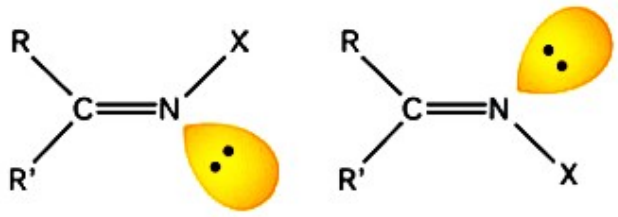


Q85 **The acid-catalyzed hydrolysis of an enamine to a carbonyl compound and a secondary amine**

Sol:

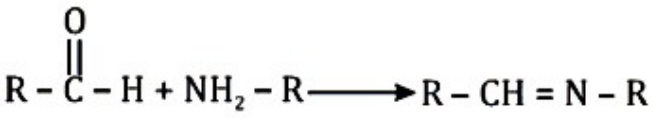


How do these mechanisms differ?
Imines can exist as stereoisomers. The isomers are named by the E, Z system of nomenclature. (The lone pair has the lowest priority.)

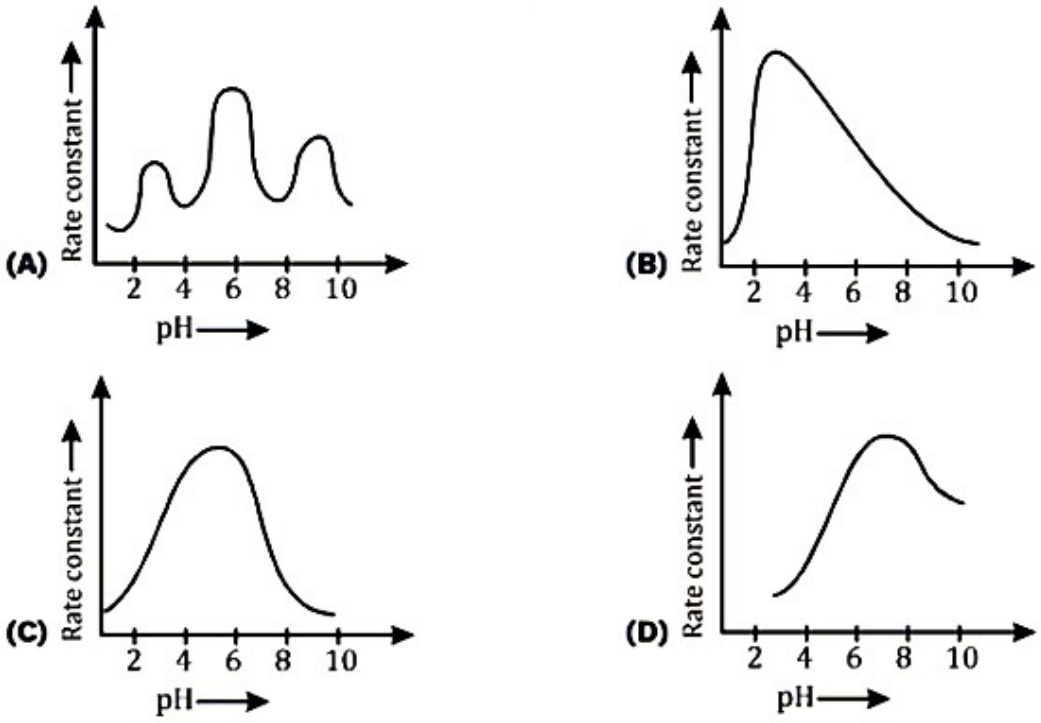


Illustrations

Q1

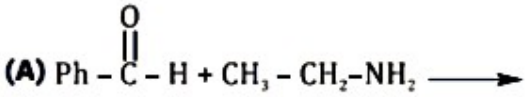


Correct rate constant versus pH diagram for above reaction.

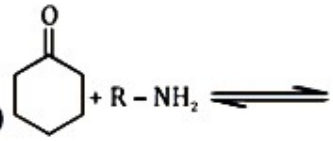


Sol: **(C)**

Q2 **In which of the following of schiff-base take place?**

(A) 

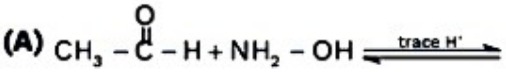
(B) 3-pentanone + benzyl amine 

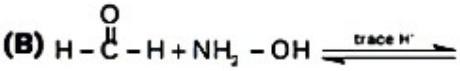
(C) 

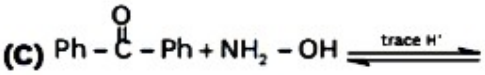
(D) All

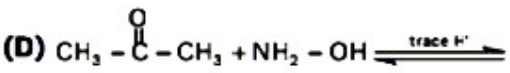
Sol: **(D)**

Q3 **In which of the reaction formation of geometrical isomer will take place.**

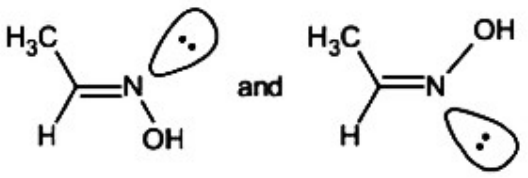
(A) 

(B) 

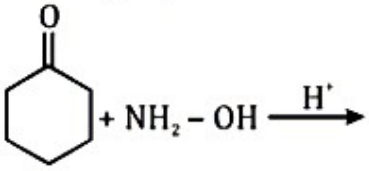
(C) 

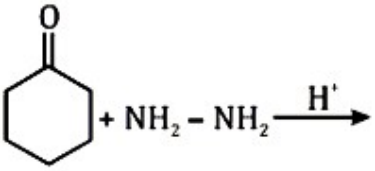
(D) 

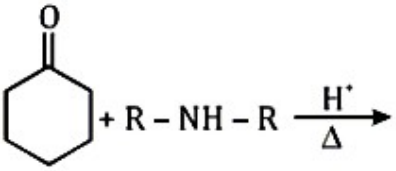
Sol: **(A)**

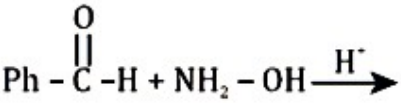


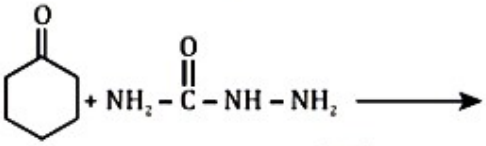
Q4 **Identify major products.**

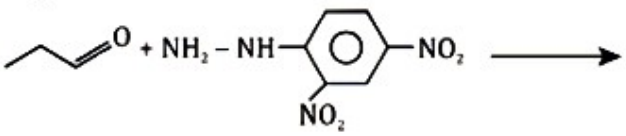
(1) 

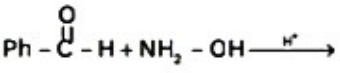
(2) 

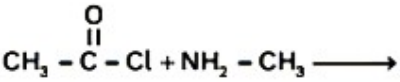
(3) 

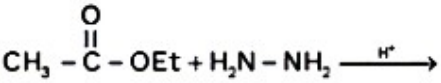
(4) 

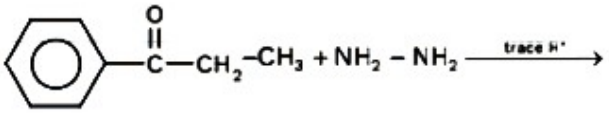
(5) 

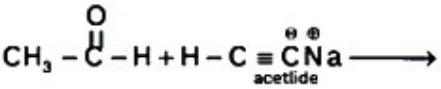
(6) 

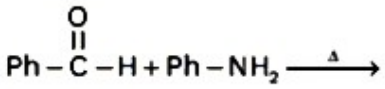
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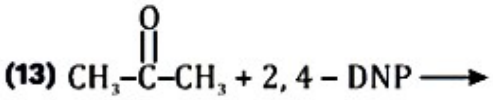
(8) 

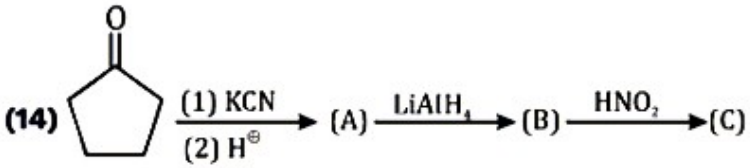
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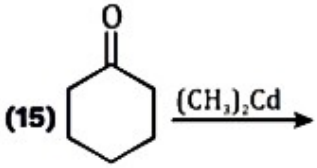
(10) 

(11) 

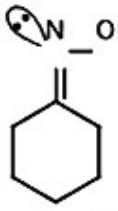
(12) 

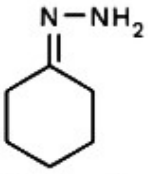
(13) 

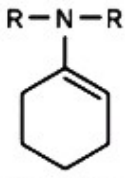
(14) 

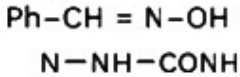
(15) 

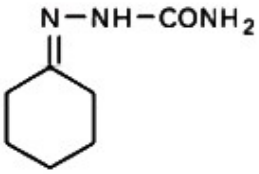
Sol:

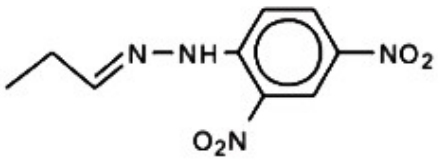
(1) 

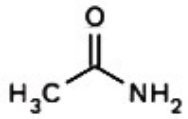
(2) 

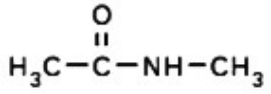
(3) 

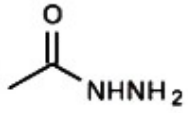
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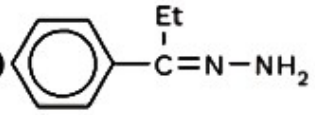
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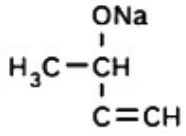
(6) 

(7) 

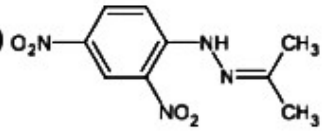
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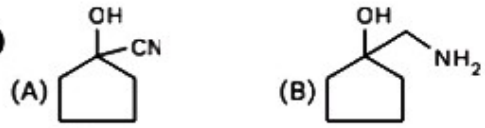
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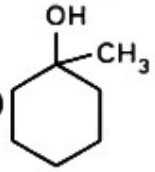
(10) 

(11) 

(12) Ph–CH=N–PH

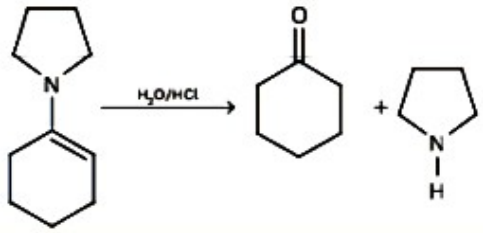
(13) 

(14) 

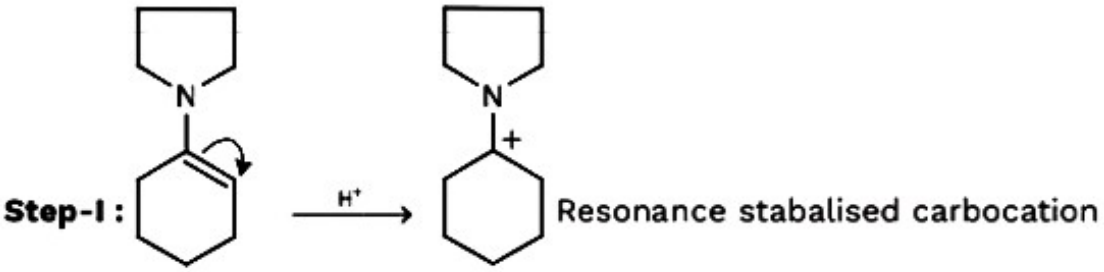
(15) 

Q5

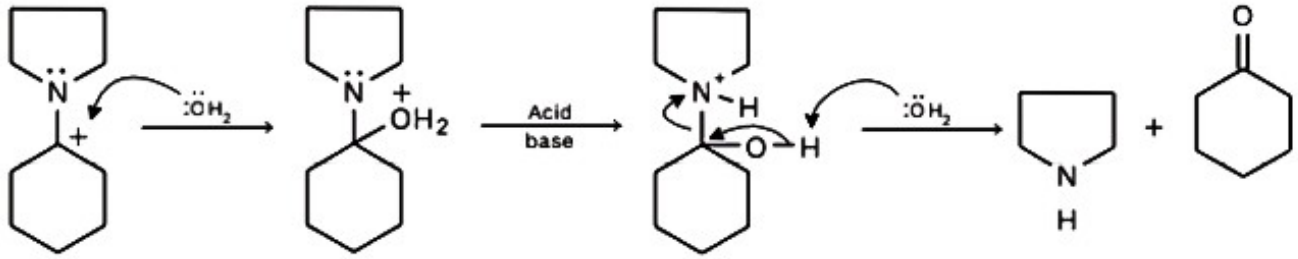
Write mechanism.

(1) 

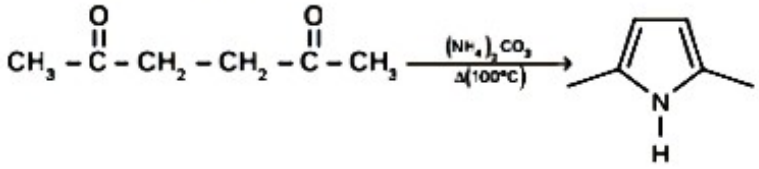
Sol:



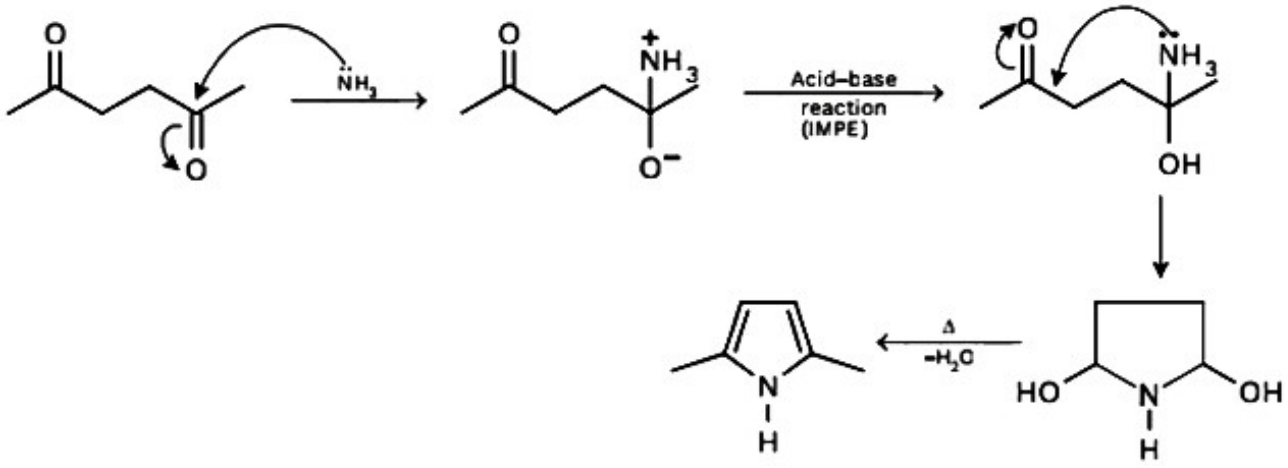
Step-II :



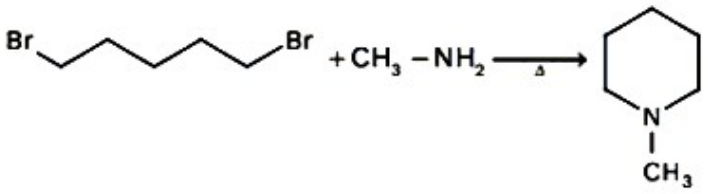
Q6 **Write mechanism.**



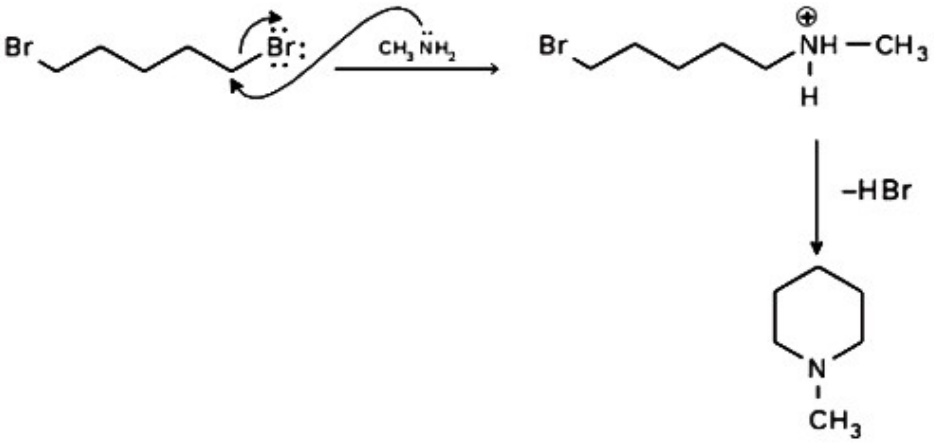
Sol:



Q7 **Write mechanism.**

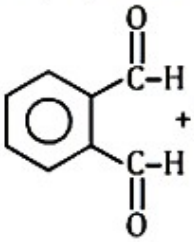


Sol:



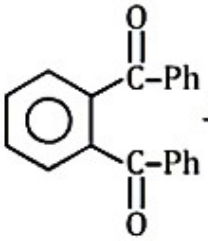
Q8 **Identify - product of addition reaction with carbonyl group**

(1)

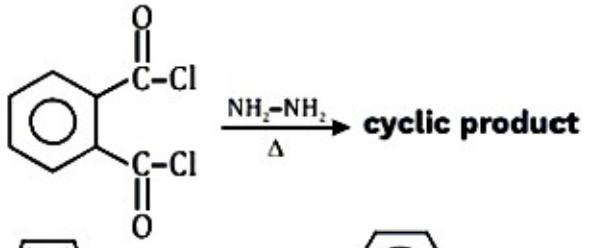


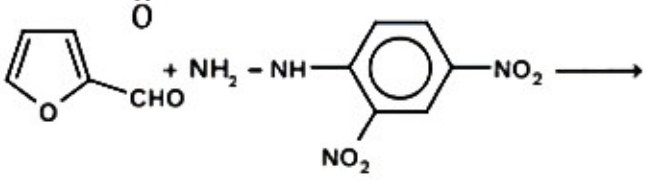
+ $NH_2–NH_2$ $\xrightarrow[\Delta]{}$ **cyclic product**

(2)

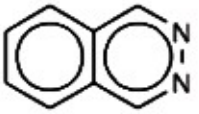


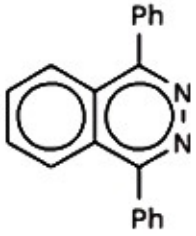
$\xrightarrow{NH_2–NH_2}$ cyclic product

(3) 

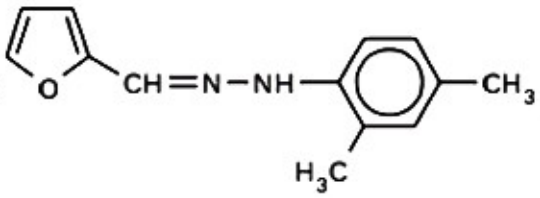
(4) 

Sol:

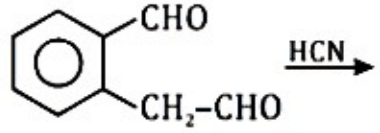
(1) 

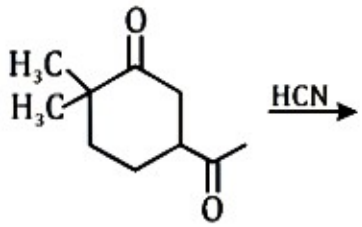
(2) 

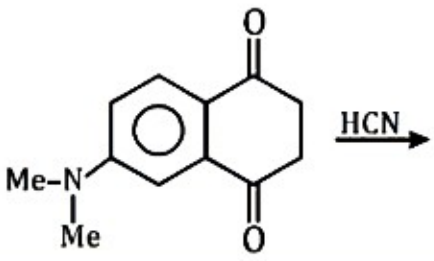
(3) 

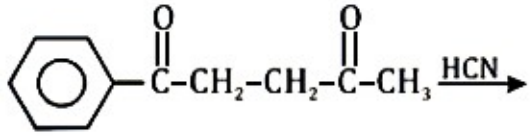
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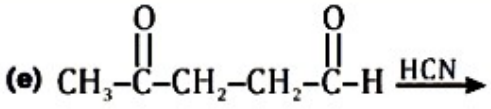
Q9 **Write chemical equation for the reaction of each of the following compounds with one equivalent of HCN.**

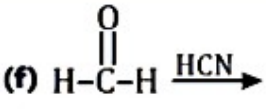
(a) 

(b) 

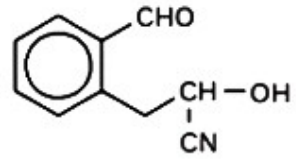
(c) 

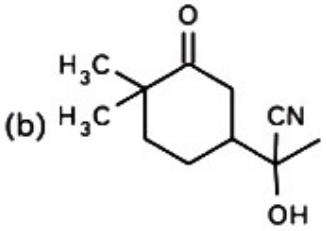
(d) 

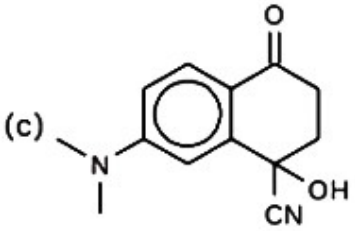
(e) 

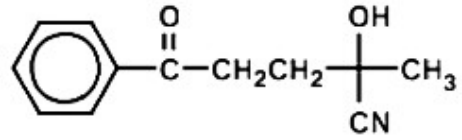
(f) 

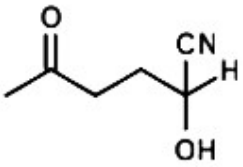
Sol:

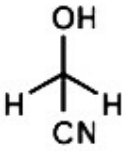
(a) 

(b) 

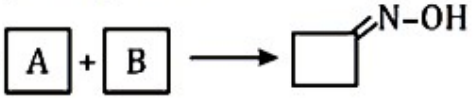
(c) 

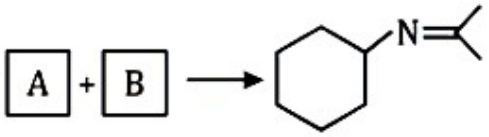
(d) 

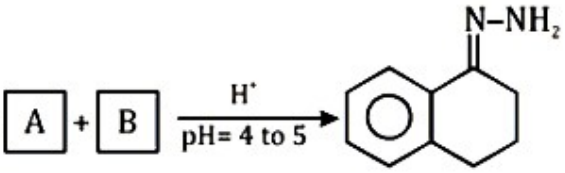
(e) 

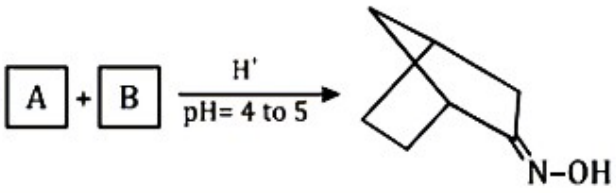
(f) 

Q10 **Show how would you prepare the following derivative from appropriate carbonyl compounds?**

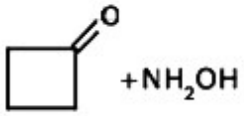
(a) 

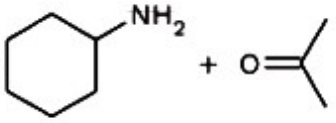
(b) 

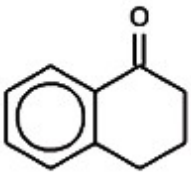
(c) 

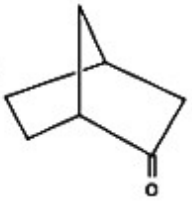
(d) 

Sol:

(a)  $+NH_2OH$

(b) 

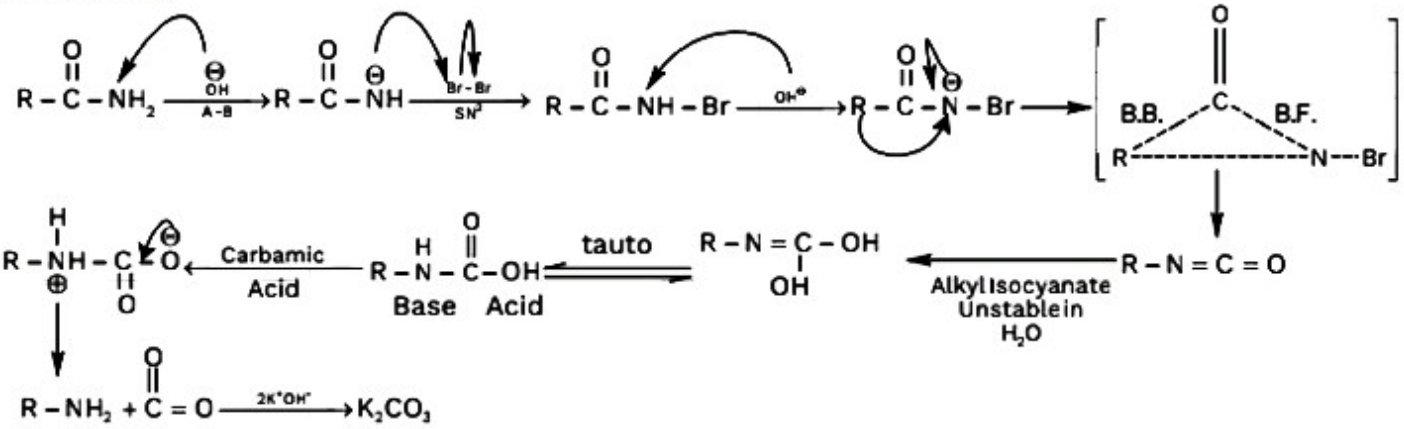
(c)  $+ NH_2NH_2$

(d)  $+ H_2N—OH$

Hoffmann Hypo Bromamide Reaction (H.H.B.)

$$R-\underset{\underset{O}{\|}}{C}-NH_2 + 4KOH + Br_2 \xrightarrow{\Delta} R-NH_2 + K_2CO_3 + 2KBr$$

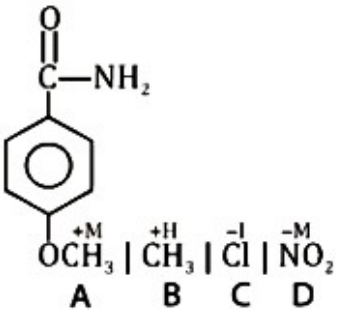
Mechanism



Some Facts:

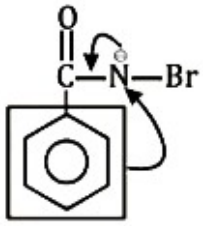
(i) In H.H.B. 1° amide inter convert into 1° amine in presence of 4 mole KOH with Br_2.

(ii) In H.H.Br, nitrene type T.S. formation is R.D.S. it means more the electron donar of more the state of reaction.



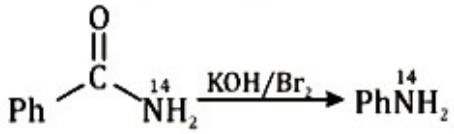
Rate : A > B > C > D

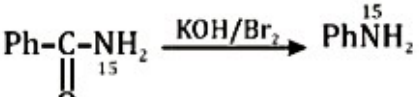
Reason :

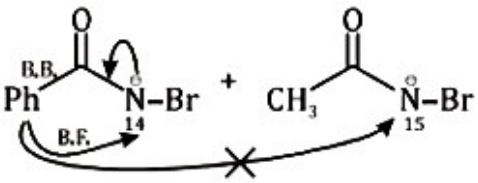


Easy to migrate due to more electron density.

(iii) HHB reaction is a intramolecular process because B.B. & B.F. both steps are simultanously it means migration is regiospecific for a reaction.

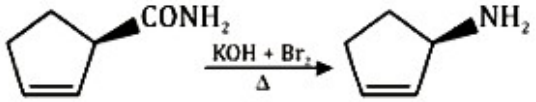






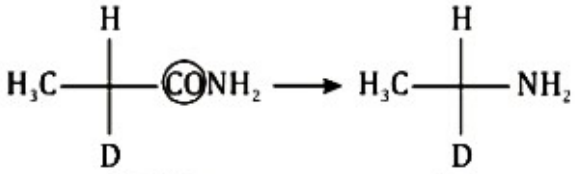
(iv) In H.H.B. reaction streochemisty is preserved, because B.B. & B.F. both steps are simultaneously.

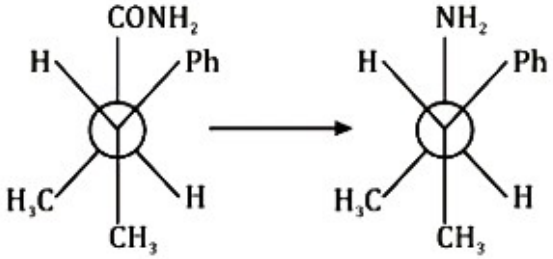
1.

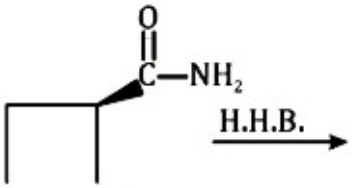


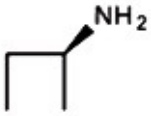
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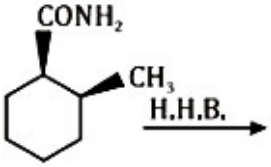


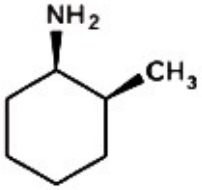
3. 

4. 

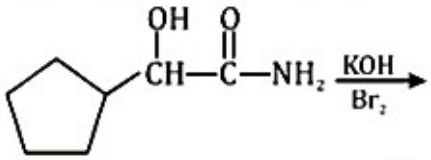
5. 

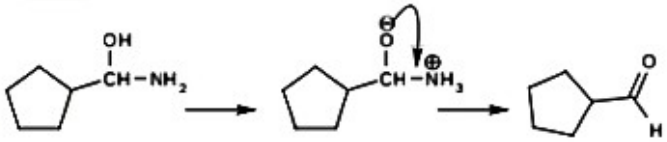
Sol: 

6. 

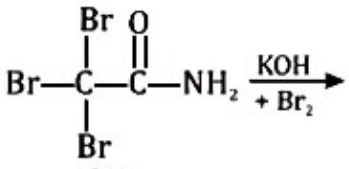
Sol: 

(V) If α -Hydroxy amide then product is further modified.

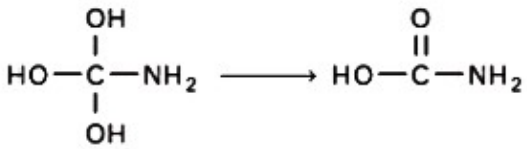
(1) 

Sol: 

(2)

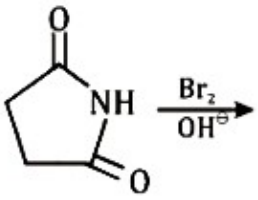


Sol:

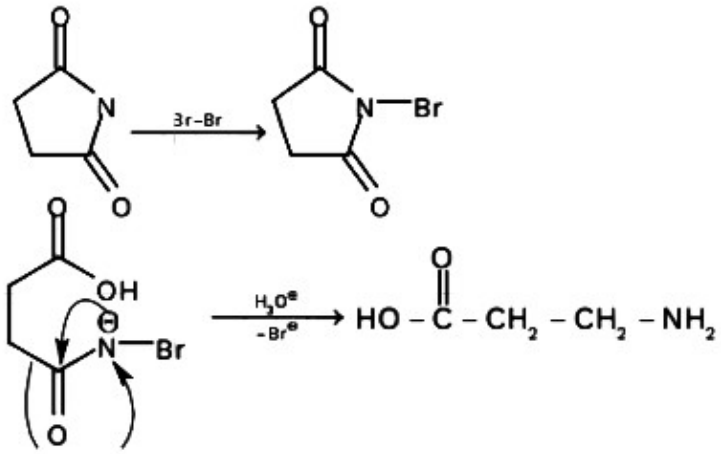


(VI) 2° Imide also give H.H.B. reaction because during reaction it converts in N-Bromo amide.

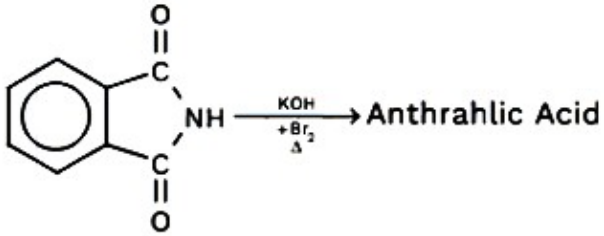
Example: 2° amide



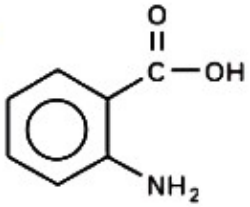
Sol:



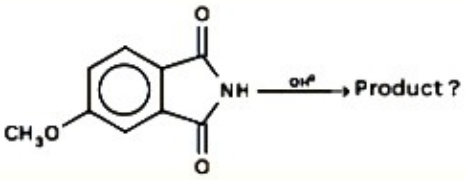
Q11



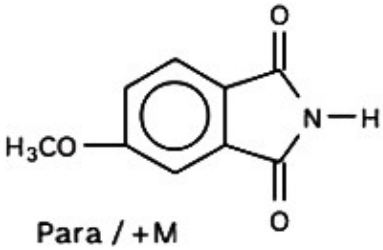
Sol:



Q12

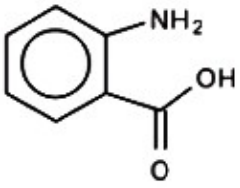


Sol:

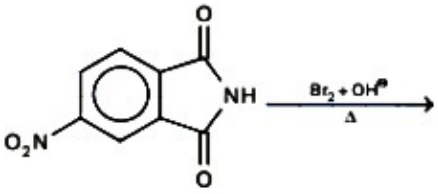


This side will migrate

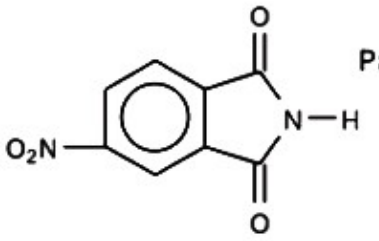
Para / +M



Q13

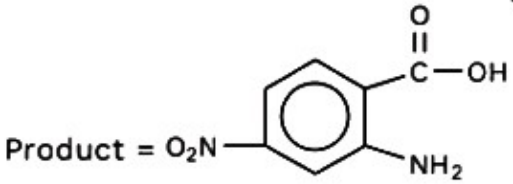


Sol:



Para side $(-M)$

This side will migrate



Q14 **Identify which compound give H.H.B. reaction?**

(A) $CH_3-\overset{\overset{O}{\|}}{C}-NH_2$ **(B)** $CH_3-\overset{\overset{O}{\|}}{C}-NH-CH_3$

(C) $CH_3-\overset{\overset{O}{\|}}{C}-N\begin{matrix}\diagup\\\diagdown\end{matrix}$ **(D)** $CH_3-\overset{\overset{O}{\|}}{C}-NH-O-\overset{\overset{O}{\|}}{C}-CH_3$

(E) $CH_3-\overset{\overset{O}{\|}}{C}-NH-Br$

Sol: **A,D,E**

(VII) In the reaction reactive base is Potassium Hypobromamide (OBr^-)

$$KOH + Br-Br \longrightarrow KBr + HOBr \xrightarrow{KOH} \overset{\ominus\ominus}{K}\overset{\ominus}{O}Br + H_2O$$

Tips and Trick

$$R-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow[\text{Same}]{OH/OBr}$$

(VIII) In Presence of different solvent, product can be modified.

Examples:

Q15

$$Ph-\overset{\overset{O}{\|}}{C}-ND_2 \xrightarrow[Br_2]{\text{(i) KOH}} \text{Product}$$

$$\downarrow \text{(i) KOH + } Br_2,\ \text{(ii) } H_2O$$

Product

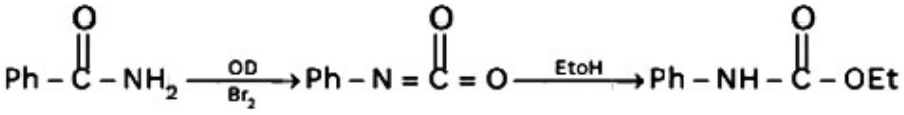
Sol:

$$Ph-\overset{\overset{O}{\|}}{C}-N\begin{matrix}D\\\diagdown\\\diagup\\D\end{matrix} \xrightarrow[\Delta OH]{Br_2} Ph-N=C=O \xrightarrow{H_2O} Ph-NH_2$$

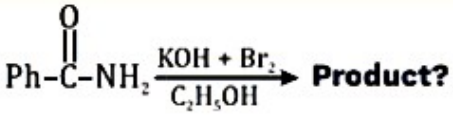
Q16

$$Ph-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow[C_2H_5OH]{KOD + Br_2} \textbf{Product?}$$

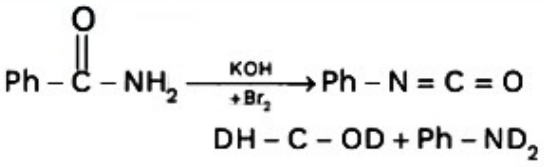
Sol:



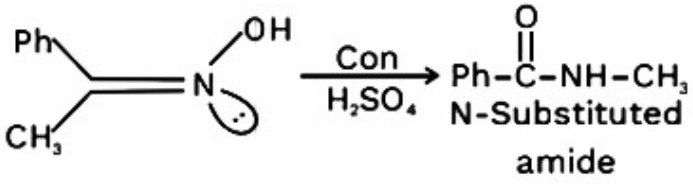
Q17



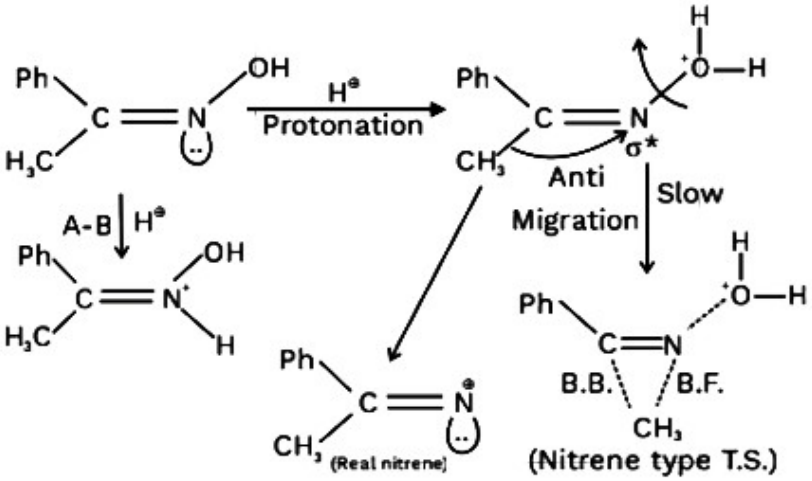
Sol:

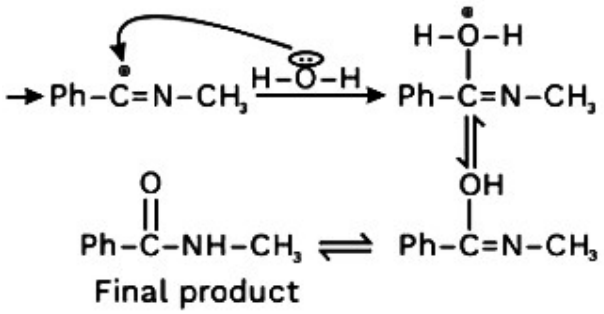


Beckmann Rearrangement

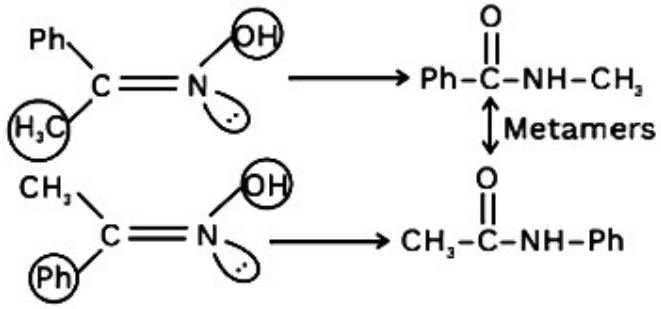


Mechanism





Trick



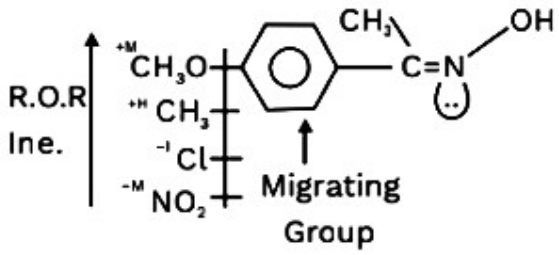
Some facts

(i) In B.M.R., oxime inter convert into N-substituted amide.

(ii) In B.M.R. anti migration takes place because of restricted rotation (p bond)

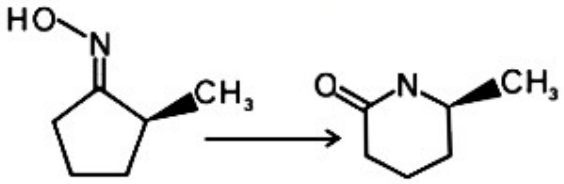
(iii) In Slow step, nitrene type T.S. formation takes place.

(iv) More the e^- donor, more the rate of reaction.

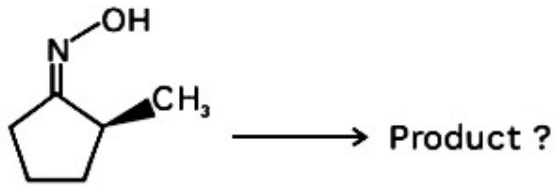


(v) In B.M.R. Stereochemistry preserved because B.F. & B.B. both steps are simultaneously

(1)



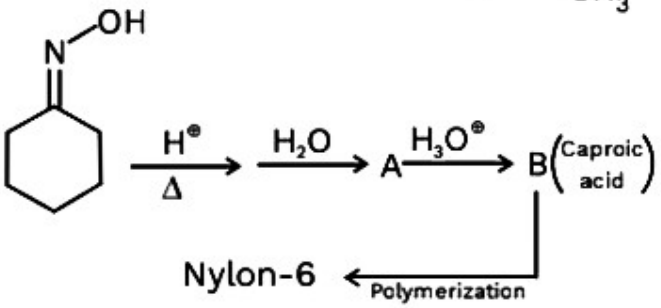
(2)



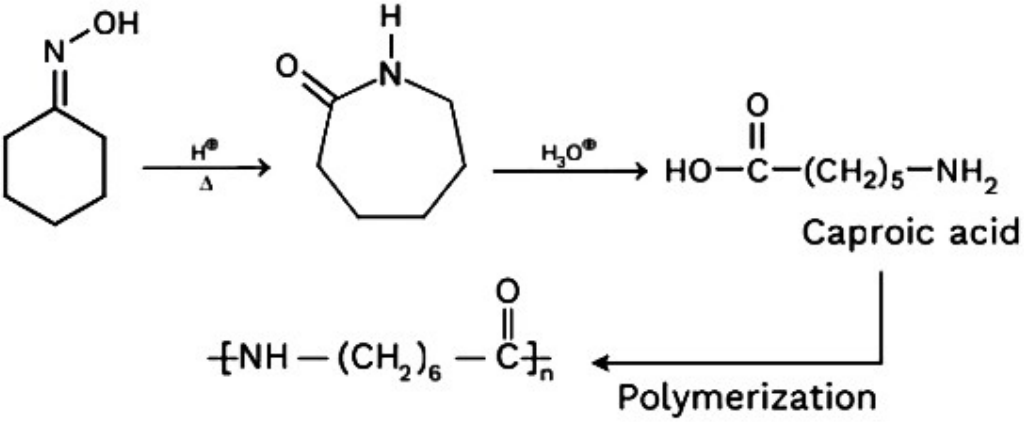
Sol:



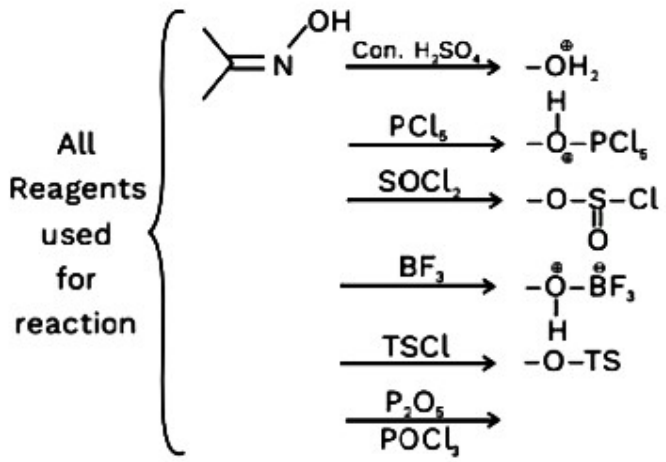
(3)

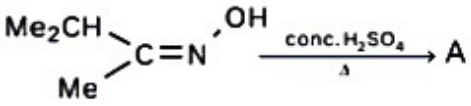


Sol:

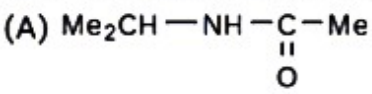


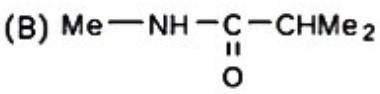
(vi) In B.M.R. different reagent can be used for the conversion of bad leaving group into good leaving group.

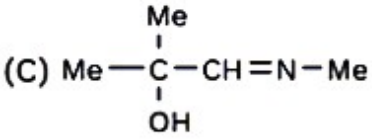


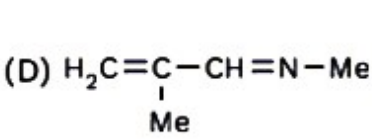
Q18 

Major product of this reaction ?

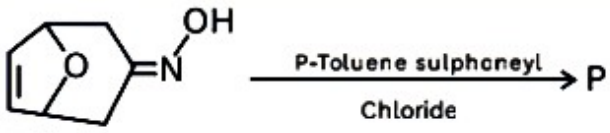
(A) 

(B) 

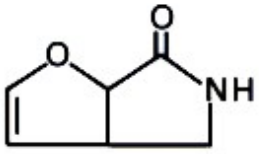
(C) 

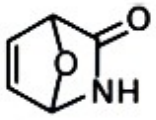
(D) 

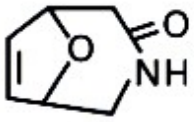
Sol: **(B)**

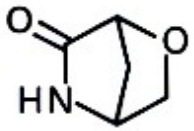
Q19 

P is :

(A) 

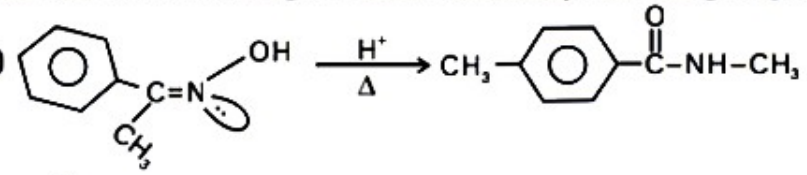
(B) 

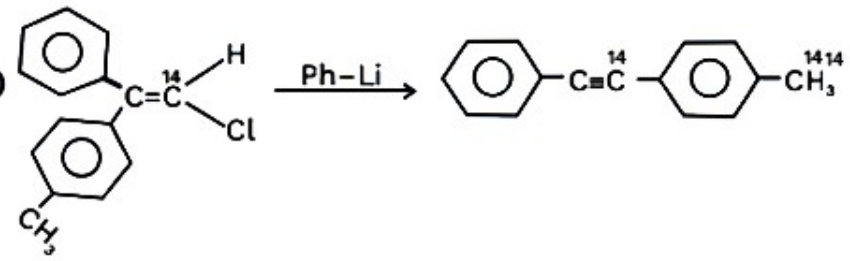
(C) 

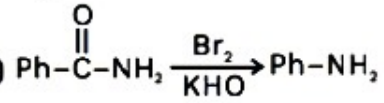
(D) 

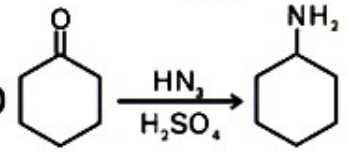
Sol: **(C)**

Q20 **Which of the following reaction is not representing major product.**

(A) 

(B) 

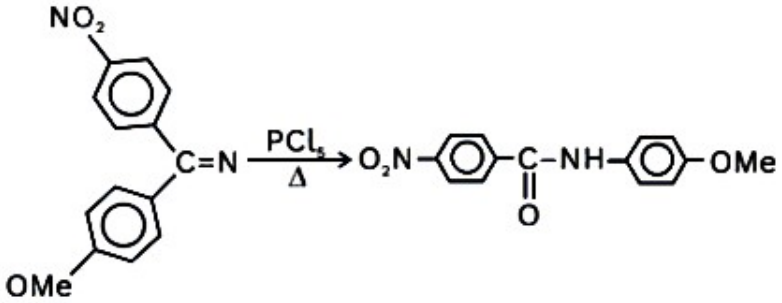
(C) 

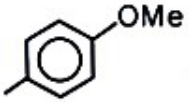
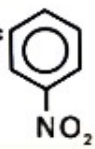
(D) 

Sol: **(BCD)**

Q21 **This question consist of two statements, printed as assertion and reason, while answering this question you are required to choose any one of the following responses.**

Assertion :

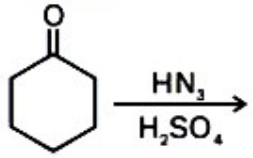


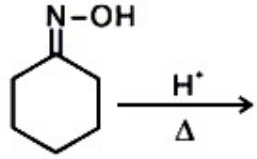
Reason : Migratory aptitude of  **group is greater than migratory aptitude of**  **group during cation rearrangements.**

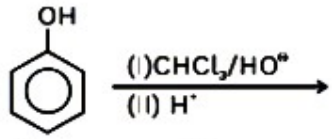
(A) If assertion is true but the reason is false
(B) If assertion is false but the reason is true
(C) If both assertion and reason are true and the reason is a correct explanation of assertion.
(D) If both assertion and reason are true but reason is not a correct explanation of assertion.

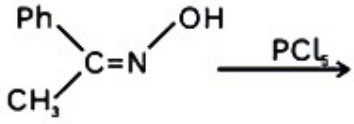
Sol: **(B)**

Q22 **Match the column**
Column (i)
(A) Caprolactum formation take place in
(B) Beckmann rearrangement is
(C) Schmidt reaction is
(D) Reaction in which number of carbon increases
Column (ii)

(P) 

(Q) 

(R) 

(S) 

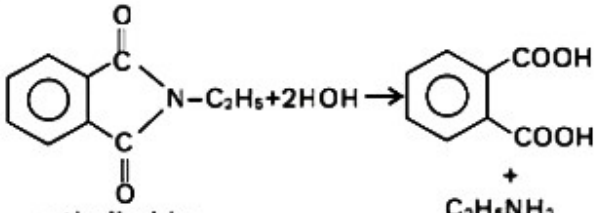
Sol: **(A) P,Q (B) Q,S (C) P (D) R**

Q23 **N-Ethyl pthalimide on hydrolysis gives–**

(A) Methyl alcohol **(B) Ethyl amine**

(C) Dimethyl amine **(D) Diethyl amine**

Sol: **[B]**



Therefore option (B) is correct.

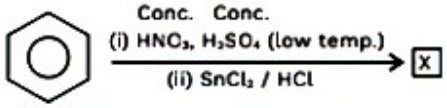
Q24 **Which of the following reaction does not give 1° amine as final product**

(A) $CH_3 - CN \xrightarrow{H_3O^{\oplus}}$ **(B)** $CH_3CN \xrightarrow{LiAlH_4}$

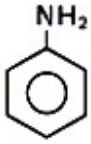
(C) $CH_3 - NC \xrightarrow{H_3O^{\oplus}}$ **(D)** $CH_3 - NC \xrightarrow{LiAlH_4}$

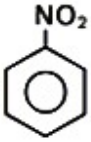
Sol: **[D]**

$CH_3 - NC \xrightarrow{LiAlH_4}$ does not give 1° amine as final product

Q25 

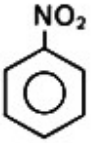
Incorrect statement about [X] is -

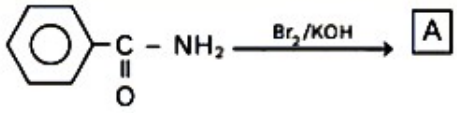
(A) X is 

(B) X is 

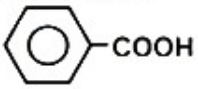
(C) X undergoes carbylamine test **(D) All of these**

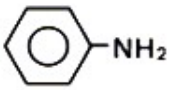
Sol: **[B]**

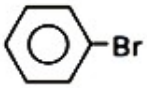
X does not 

Q26 

product (A) is

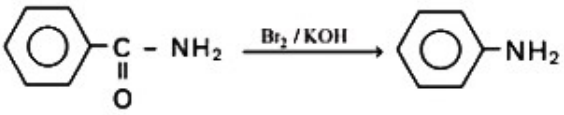
(A) 

(B) 

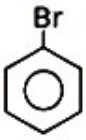
(C) 

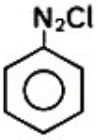
(D) None of these

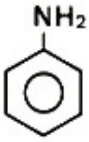
Sol: **[B]**



Q27 **Q.5**  (i) HNO_3, H_2SO_4 (ii) $SnCl_2$ / HCl (iii) HNO_2 (iv) CuBr/Δ **(P), final product P is**

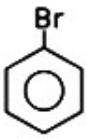
(A) 

(B) 

(C) 

(D) None of these

Sol: **[A]**



Q28 **The correct set of the products obtained in the following reactions–**

(a) $RCN \xrightarrow[\text{hydrolysis}]{\text{reduction}}$ **(b)** $RCN \xrightarrow[\text{(ii) } H_2O]{\text{(i) } CH_3MgBr}$

(c) $RNC \xrightarrow{}$ **(d)** $RNH_2 \xrightarrow{HNO_2}$

The answer is–

(A)	**A**	**B**
	2° Amine	**Methyl ketone**
	C	**D**
	1° Amine	**Alcohol**
(B)	**A**	**B**
	1° Amine	**Methyl ketone**
	C	**D**
	1° Amine	**Alcohol**
(C)	**A**	**B**
	2° Amine	**Methyl ketone**
	C	**D**
	2° Amine	**Acid**
(D)	**A**	**B**
	2° Amine	**Methyl ketone**
	C	**D**
	2° Amine	**Aldehyde**

Sol: **[B]**

(a) $R-CN \xrightarrow{\text{reduction}} R-CH_2NH_2$

1° Amine

(b) $R-CN \xrightarrow[H_2O]{CH_3MgBr} R-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}= N-MgBr \xrightarrow{H_2O} R-\underset{\displaystyle O}{\overset{\displaystyle ||}{C}}-CH_3 \quad NH_3 + Mg\begin{matrix}Br\\OH\end{matrix}$

Methyl ketone

(c) $R-NC \xrightarrow{\text{hydrolysis}} R-NH-CH_3$

2° Amine

(d) $R-NH_2 \xrightarrow{H_2O} R-OH$

Alcohol

Therefore option (B) is correct.

Q29 **Which amine will not react with nitrous acid–**

(A) Methyl amine **(B) Ethyl amine**

(C) Dimethyl amine **(D) N, N Dimethyl ethane amine**

Sol: **[D]**

Due to steric hindrance N, N-dimethyl ethane amine can not be reset with nitrous acid.

Therefore option (D) is correct

Q30 **Which of the following compound cannot be produced if 1-propane amine is treated with $NaNO_2$ and HCl –**

(A) Propane -1-ol **(B) Propane-2-ol**

(C) 2-Chloropropane **(D) 2-Propaneamine**

Sol: **[D]**

$$CH_3CH_2CH_2-NH_2 + NaNO_2 + 2HCl \xrightarrow[H_2O]{(HONO)}$$

$[CH_3CH_2CH_2^+ - N\equiv N:Cl^-]$

diazonium salt (highly unstable) + NaCl

$\downarrow -N_2$

$[CH_3CH_2\dot{C}H_2 + Cl^-] \leftrightarrow [CH_3-\dot{C}H_2-CH_3]$

$\rightarrow$ Alkene $CH_3CH=CH_2$ + Alcohol $[CH_3CH_2CH_2-OH$ or $CH_3-\underset{OH}{CH}-CH_3]$ + Alkyl halides $[CH_3CH_2CH_2-Cl$ or $CH_3-\underset{Cl}{CH}-CH_3]$

Therefore option (D) is correct

Q31 **Which of the following amine does not react with Hinsberg reagent–**

(A) Neopentyl amine **(B) Isopropyl amine**

(C) Triethyl amine **(D) Ethyl methyl amine**

Sol: **[C]**

Only 1° and 2° amine react Hinsberg reagent. 3° amine can not be react with Hinsberg reagent. Therefore triethyl amine $(C_2H_5)_3\ddot{N}$. Which is 3° Amine can't be react with Hinsberg reagent.

option (C) is correct

Q32 **Methyl amine on reaction with chloroform in the presence of NaOH gives–**

(A) Methyl isocyanide **(B) Methyl chloride**

(C) N-Methylchloramine **(D) Chloramine**

Sol: $CH_3NH_2 + Cl_2 \xrightarrow{NaOH} CH_3-NH-Cl + HCl$

N-methyl chloroamine

There fore option (C) is correct.

Q33 **Which of the following diazonium salt is relatively stable of 0-5°C–**

(A) $CH_3-N\equiv N\}^{\oplus}Cl^-$ **(B)** $CH_3-C(CH_3)_2-N\equiv N\}^{\oplus}Cl^-$

(C) $C_6H_5-N\equiv N\}^{\oplus}Cl^-$ **(D)** $(CH_3)_3C-N\equiv N\}^{\oplus}Cl^-$

Sol: **[C]**

Benzene diazonium salt is stable at 0–5°C due to the resonance.

Therefore option (C) is correct.

Q34 **Alkylamine dissolve in hydrochloric acid to form alkylammonium chloride. The nitrogen in the latter salt is–**

(A) Quadricovalent only

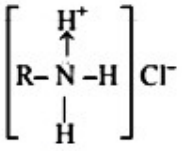
(B) Tricovalent only

(C) Unielectrovalent only

(D) Quadricovalent, Unielectrovalent

Sol: **[D]**

$R-NH_2 + HCl \longrightarrow [\ RNH_3^+\ ]Cl^-$ or



alkylammonium chloride

In this salt nitrogen have Quadricovalent and Unielectrovalent bond.

There fore option (D) is correct.

Q35 **The presence of primary amines can be confirmed by–**

(A) Reaction with HNO_2

(B) Reaction with $CHCl_3$ and alc. KOH

(C) Reaction with Grignard reagent

(D) Reaction with acetyl chloride

Sol: **[B]**

Only primary amine give carbyl amine reaction

$R-NH_2 + CHCl_3 \xrightarrow{KOH} R-\overset{+}{N} \equiv \bar{C} + 3KCl + 3H_2O$

There fore option (B) is correct.

Q36 **In the reaction $CH_3CH_2NH_2 + CH_3MgBr \rightarrow X$, the product is–**

(A) CH_3CH_3 **(B) $CH_3CH_2CH_3$**

(C) $CH_3CH_2CH_2CH_3$ **(D) CH_4**

Sol: **[D]**

$$CH_3-CH_2NH_2+CH_3MgBr \rightarrow \underset{\text{Methane}}{CH_4} + Mg\begin{matrix}Br \\ NH-CH_2CH_3\end{matrix}$$

Therefore option (D) is correct.

Q37 **The compound obtained by the reaction between primary amine and aldehyde is–**

(A) An amide **(B) imine** **(C) Nitrite** **(D) Nitro**

Sol: **[B]**

$$R-CH=\boxed{O+\begin{matrix}H \\ H\end{matrix}}N-R \rightarrow \underset{\text{aldimin}}{R-CH=N}$$

Therefore option (B) is correct

Q38 **Name the products in the acid- base reaction-**

(I) $CH_3CH_2NH_2 + HI$ (II) $(CH_3)_3N + HBr$

(A) (I) Trimethyl ammonium iodide (II) Trimethyl ammonium bromide

(B) (I) Ethyl ammonium iodide (II) Methyl ammonium bromide

(C) (I) Ethyl ammonium iodide (II) Trimethyl ammonium bromide

(D) All of these

Sol: **[C]**

$C_2H_5NH_2 + HI \rightarrow C_2H_5NH-I$

Ethyl ammonium iodide

$(CH_3)_3N + HBr \rightarrow (CH_3)_3N - Br$

trimethyl ammonium bromide

Therefore option (C) is correct.

Q39 **When propionamide reacts with Br_2 in the presence of alkali the product is–**

(A) $CH_3CH_2CH_2NH_2$ **(B)** $CH_3CH_2NH_2$

(C) C_3H_7CN **(D)** C_2H_5CN

Sol: **[B]**

$$CH_3-CH_2-\underset{\underset{O}{\|}}{C}-NH_2 \xrightarrow[Br_2]{KOH} CH_3CH_2-NH_2+KBr$$

There fore option (B) is correct.

Q40 **A reaction of ethyl amine & acetic anhydride leads to the formation of–**

(A) $CH_3NHCOCH_3$ **(B)** $C_2H_5CONHCH_3$

(C) $CH_3CONHC_2H_5$ **(D)** $CH_3-CH=N-OC_2H_5$

Sol: **[C]**

$$CH_3CH_2-NH_2 + (CH_3CO)_2O \longrightarrow CH_3-\underset{\underset{O}{\|}}{C}-NH-CH_2-CH_3+CH_3COOH$$

Therefore option (C) is correct.

Q41 **Which of the following is not correct–**

(A) Ethyl amine & aniline both have NH_2 group

(B) Ethyl amine & aniline both dissolve in HCl

(C) Ethyl amine & aniline both react with $CHCl_3$ & KOH to form unpleasant gas

(D) Ethyl amine and aniline both react with HNO_2 to give hydroxy compound

Sol: **[D]**

Ethyl amine react with HNO_2 ($NaNO_2$+HCl) and formed ethyl alcohol

$$C_2H_5NH_2 \xrightarrow{HNO_2} C_2H_5OH$$

But $C_6H_5-NH_2$ react with HNO_2($NaNO_2$+HCl) to give benzene diazonium salt (BDC)

$$C_6H_5NH_2 \xrightarrow{NaNO_2+HCl} \underset{BDC}{C_6H_5N^+Cl^-}$$

Therefore option (D) is correct.

Q42 **Cyanides exists in–**

(A) Tautomeric form **(B) Geometrical form**

(C) In both form **(D) None**

Sol: **[A]**

Cyanides exists in tautomeric form.



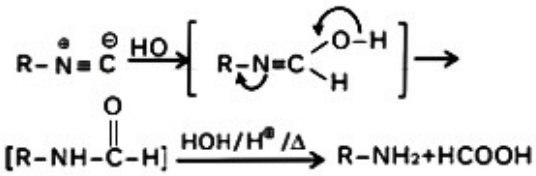
Therefore option (A) is correct.

Q43 **Hydrolysis of alkyl isocyanide yields–**

(A) Primary amine **(B) Tert. amine**

(C) Alcohol **(D) Aldehyde**

Sol: **[A]**



So the primary amine is formed by hydrolysis of alkyl isocyanide

Therefore option (A) is correct.

Q44 **Dipole moment of which of the following is highest-**

(A) Ether **(B) Alcohol**

(C) Halo alkane **(D) Nitro alkane**

Sol: **[D]**

Factual

Q45 $CH_3-CH_2-Cl \xrightarrow{NaNO_2} A$; $CH_3-\underset{Cl}{\underset{|}{CH_2}} \xrightarrow{AgNO_2} B$. **A and B are–**

(A) $CH_3-CH_2-NO_2$, $CH_3-CH_2-O-N=O$

(B) $CH_3-CH_2-O-N=O$, $CH_3-CH_2-NO_2$

(C) Both are $CH_3-CH_2-NO_2$

(D) Both are $CH_3-\underset{NO}{\underset{|}{CH_2}}$

Sol: **[B]**

$CH_3-CH_2-O-N=O$, $CH_3-CH_2-NO_2$

Q46 $CH_3-CH_2-\underset{NO}{\underset{|}{CH_2}} \xrightarrow[NaOH]{Br_2} A$

Here A is–

(A) $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-OH$

(B) $CH_3-CH_2-\underset{Br}{\underset{|}{CH}}-NO_2$

(C) $CH_3-\underset{Br}{\underset{|}{CH}}-CH_2-NO_2$

(D) $\underset{Br}{\underset{|}{CH_2}}-CH_2-CH_2-NO_2$

Sol: **[B]**

$CH_3-CH_2-\underset{Br}{\underset{|}{CH}}-NO_2$

Q47 **Which of the following is functional isomer of nitro methane–**

(A) CH_3-NO_2

(B) CH_3-O-NH_2

(C) $CH_3-O-N=O$

(D) $O-\underset{CH_3}{\underset{|}{N}}=O$

Sol: **[C]**

$CH_3-O-N=O$ is functional isomer of nitro methane